EXACT SOURCE IDENTITIES AND NORMALIZED REFERENCE RESPONSE FOR LATTICE QED IN THREE DIMENSIONS

Abstract. We prove a cutoff-uniform estimate for the normalized external response of three-dimensional lattice QED on its reference torus. The mass and vacuum parameters are those of one common tuned trajectory. An exact unfolding identifies the zero-flux law with a harmonic-averaged Gaussian integral, and a source shift expresses Wick products as determinant derivatives. We compare the terminal logarithm with its one-cell free factors, estimate connected derivatives between separated cells, and control the complex large-field complement. A finite fourth-order cancellation then gives a positive power of the charge, uniformly in the ultraviolet cutoff, on a nonempty interval of charges.

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1. Introduction

We use the chamber regulator with weights $(6, 6\sqrt{2}, 6\sqrt{2}, 6)$ and Wilson parameter $0 < \rho < 1$. The operator $D^{\mathrm{ch}}[B]$ acts on four-component spinors with 48 chamber labels per site, antiperiodic in every coordinate; its argument B is the bond phase. The stencil is recalled in section 1.1.

Fix the block construction of [26g, Proposition 4.1 and §4.1–4.4], with its positive charge ceilings $e_1 \le e_0 < 1$ and the common-pair theorem [26a, Theorem D]. Thus $L \ge 13$ is odd, $M = L^m$ with $m \in \mathbb{Z}_{\ge 2}$, $0 < \alpha < 1$, $1 - \alpha < \varepsilon \le 1/200$, and the localization and cutoff powers are $r_{\mathrm{loc}} = 2$, $p_0 = 9$, $p = 19$. The fixed ceilings denoted by e_0, e_1 here are the $\bar{e}_0, \bar{e}_1$ of [26g]; they are distinct from the running charge e_j at $j = 0$. These fixed constants include the photon and fermion inverse bounds, the exact block step and the terminal conditions. The parameter input is Proposition 4.1 of [26g], which supplies the construction and charge ceilings before any probe-dependent restriction. The response estimates are proved here.

For $0 < \bar{m} \le 1$ put

$$h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}, \qquad \eta = L^{-N}, \qquad q = e\sqrt{\eta}, \qquad K = N + \log_L h.$$

Let r_K be the least nonnegative integer power of L at least $[-\log(e\sqrt{h})]^2$, and set

$$R_{\mathrm{top}} = hMr_K, \qquad R^* = R_{\mathrm{top}}/L, \qquad n_* = R^*/\eta, \qquad \ell_c = R^*/(2L), \qquad N_* = 8L^3.$$

The finite threshold $N_{\mathrm{ref}}(e)$ and stopping level on R^* are those of [26a, Proposition 6.1 and (53)]. For $N \geq N_{\mathrm{ref}}(e)$ use the real pair $P_N(e) = (m_{0,N}, \varepsilon^{\mathrm{v}}_{0,N})$ selected on R_{top} by [26a, Theorem D], and write $\boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}$. This same pair is used for every source and every twist. Its vacuum factor cancels from the normalized response below.

Fix a real nonzero $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, put $f = dd^*\beta$, and let D_β be the side of a least coordinate cube containing $\operatorname{supp}\beta$. Enlarge that cube by two in each direction to obtain Q_β of side $D_\beta^{\mathrm{buf}} = D_\beta + 4$ and center z_β. With J_η denoting cube averaging and $\delta_{\eta,\mu}$ the centered difference, define the covering bond cochain

$$\widehat{a}^o_\nu(x) = -\eta^{1/2} \sum_{\mu \neq \nu} (\delta_{\eta,\mu} J_\eta \beta_{\mu\nu})(\eta x + \eta e_\nu/2). \tag{1}$$

A placement $v \in \eta\mathbb{Z}^3$ is admitted when $v + Q_\beta \subset [\ell_c/8, 7\ell_c/8]^3$. Translate the probe by v, reflect it into the cells indexed by $\mathcal{C}_* = (\mathbb{Z}/2L\mathbb{Z})^3$, and periodize on $C(n_*)$. The resulting cochains are $\widehat{a}_c$; the reflection maps and their orientation action are specified in section 2. They have mutually orthogonal curvatures, all of squared norm $v_\eta = \|d\widehat{a}^o\|^2_{\ell^2}$.

Write $C(n_*) = \mathbb{Z}^3/n_*\mathbb{Z}^3$ with its oriented cubical cells. Let γ_{n_*} be the standard Gaussian on its exact real two-cochains, with counting inner product. For such a curvature u, let $A(u)$ be its mean-zero transverse potential. Set $H_{\theta,\nu} = \theta_\nu/(qn_*)$ and $a(t) = e^{-1}\sum_c t_c \widehat{a}_c$. Define

$$\mathring{\Phi}(t) = \frac{\displaystyle\int_{[-\pi,\pi]^3} \frac{\mathrm{d}\theta}{(2\pi)^3} \int \det\bigl(D^{\mathrm{ch}}[q(A(u) + H_\theta + a(t))] + \boldsymbol{\mu}_N\bigr)\, \mathrm{d}\gamma_{n_*}(u)}{\displaystyle\int_{[-\pi,\pi]^3} \frac{\mathrm{d}\theta}{(2\pi)^3} \int \det\bigl(D^{\mathrm{ch}}[q(A(u) + H_\theta)] + \boldsymbol{\mu}_N\bigr)\, \mathrm{d}\gamma_{n_*}(u)}. \tag{2}$$

The denominator is positive, including when the tuned total mass has either sign. The free one-cell ratio uses the positive reference mass:

$$N_0(t) = \frac{\det\bigl(D^{\mathrm{ch}}[q(t/e)\widehat{a}_0] + \bar{m}\eta\bigr)}{\det\bigl(D^{\mathrm{ch}}[0] + \bar{m}\eta\bigr)}, \qquad c_m = N_0^{(m)}(0), \quad m = 2, 4. \tag{3}$$

These are ordinary derivatives of the ratio. Put

$$\Theta_m^{(0)} = \left| \left[\prod_{c \in \mathcal{C}_*} (\partial_{t_c}^m - c_m) \right] \mathring{\Phi}(0) \right|^{1/N_*}, \qquad \mathfrak{e} = \tfrac{1}{2}(1/4 - 9\varepsilon) - 12\varepsilon > 0.$$

Theorem A (Normalized reference response)**.** For the fixed construction, mass $0 < \bar{m} \leq 1$, and probe β above, there is $e_{\mathrm{RR}} > 0$ such that for every $0 < e \leq e_{\mathrm{RR}}$ and every integer $N \geq N_{\mathrm{ref}}(e)$, the function (2) is entire and

$$\Theta_4^{(0)} \leq Ce^{\mathfrak{e}}, \qquad (1 + |c_2|)\Theta_2^{(0)} \leq Ce^{\mathfrak{e}}. \tag{4}$$

The constants are independent of N and of the admitted placement. Every centered placement, $|v + z_\beta - (\ell_c/2)(1,1,1)|_\infty \leq \eta/2$, is admitted in this range. One may take $C = 3 \cdot 10^5 (1 + \sqrt{h}\, C_\beta)^4 h^{1/8}$, where the finite, charge-independent C_β is defined in (58).

Let $\mathbb{E}^{(0)}$ denote the normalized zero-flux curvature law on this same reference torus and at the same tuned mass. Equivalently it is the probability law of $F^{\mathrm{u}} = u$ obtained by weighting γ_{n_*} with the zero-source twist-averaged determinant in (2). For $X_c = \langle d\widehat{a}_c, F^{\mathrm{u}} \rangle$ and $H_2(y) = y^2 - v_\eta$, $H_4(y) = y^4 - 6v_\eta y^2 + 3v_\eta^2$,

$$\mathbb{E}^{(0)} \prod_{c \in \mathcal{C}_*} (e^{-m} H_m(X_c) - c_m) = \left[\prod_{c \in \mathcal{C}_*} (\partial_{t_c}^m - c_m) \right] \mathring{\Phi}(0), \qquad m = 2, 4. \tag{5}$$

All expectations are finite. The response estimate concerns the full fixed family of $N_* = 8L^3$ cells and the harmonic-averaged zero-flux law. ■□

The proof has two distinct parts. The exact identities require no renormalization estimates. Integer cochain unfolding retains the gauge and harmonic Jacobians; a real source shift, followed by analytic continuation, then gives (5). For the quantitative bound, the terminal density retains the external connection throughout the common trajectory. Its logarithm is separated into the free one-cell logarithms, their errors, and terms involving several cells. A tree expansion preserves the distance between marked cells and bounds the latter terms exponentially in their separation.

The restricted terminal density defines a positive measure of mass at most one. The omitted configurations have small probability under the zero-flux Gaussian bound. A two-constants estimate transfers this smallness to the complex coefficient norm needed for differentiation. In the ring retaining degree at most four in each source, every error family which misses a cell is annihilated by that cell's free factor. The surviving families therefore carry a power of the error for every cell. This proves the estimate after taking the N_*th root. The final section verifies all additional response inequalities on one positive charge interval, including at jumps of the rounded scales.

These are the source and response arguments of [26i, §19–20, §21.1, §23]. The terminal operator estimates and exact Borel density used below are proved in [26j, Theorem B, §6 and Appendix A]. The free ratio has the convention of [26d, Theorem B]; its coefficient estimates at the analytic radius required here are proved in section 7. No sign condition on the continuum fourth coefficient is needed. Transfer to larger tori and other flux sectors is a subsequent result.

1.1. **Regulator and elementary conventions.** Let σ_μ be the Pauli matrices and $\gamma_\mu = \sigma_3 \otimes \sigma_\mu$. The labels $g \in O_{\mathrm{cub}} = O(3) \cap \mathrm{GL}(3,\mathbb{Z})$ are the signed permutation matrices. Let s_0 reverse coordinate three, s_1, s_2 exchange $(2,3)$ and $(1,2)$, and let r_{e_1} reverse coordinate one. Put

$$n_0 = e_3, \quad n_1 = (e_2 - e_3)/\sqrt{2}, \quad n_2 = (e_1 - e_2)/\sqrt{2}, \quad n_3 = -e_1, \qquad M_i(g) = \tfrac{1}{2} w_i(\rho + \gamma \cdot g n_i).$$

On a unit torus the chamber operator is

$$\begin{aligned}(D^{\mathrm{ch}}[B]\psi)(c,g) = \sum_{i=0}^{2} M_i(g)\big(\psi(c,g) - \psi(c,gs_i)\big) \\ + M_3(g)\big(\psi(c,g) - \mathrm{e}^{iB(c,c+ge_1)}\psi(c+ge_1, gr_{e_1})\big).\end{aligned} \tag{6}$$

Reversed bonds carry the negative cochain value. We write $D_A = D^{\mathrm{ch}}[qA]$ in the unit-lattice source identities. At mesh η_j, the external hop and the bond phase have the corresponding factors η_j^{-1} and $e_j\eta_j$. The spinor pairing has weight $1/48$ per chamber label; its constant weight does not change a finite-dimensional determinant.

For smooth functions,

$$J_\eta v(y) = \eta^{-3} \int_{[-\eta/2,\eta/2]^3} v(y+z)\,\mathrm{d}z, \qquad \delta_{\eta,\mu} v(y) = \frac{v(y+\eta e_\mu/2) - v(y-\eta e_\mu/2)}{\eta}.$$

We use $\tau_v f(x) = f(x-v)$ and the seminorm $|\beta|_{C^j} = \max_{\mu<\nu,|\alpha|\le j} \|\partial^\alpha \beta_{\mu\nu}\|_\infty$. Counting cochain pairings are bilinear after complexification. Distances are periodic sup distances. The symbol e is the physical charge; numerical exponentials are written $\exp(\cdot)$ or e.

2. Probes on the reference cells

The sampling and placement arguments are those of [26i, §19]. The external-source identity uses a finitely supported bond cochain. The continuum cumulant uses the test $f = dd^*\beta$. We identify the discrete source with a nearby smooth test and place it strictly inside one reference cell. Copies in distinct cells then have orthogonal curvatures, which makes the normal-ordering factor separate over cells.

2.1. **The discrete source and the smooth test.** Fix a real nonzero $\beta \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$. Use the unbuffered support-cube side D_β from the introduction, and write $D_\beta^{\mathrm{buf}} = D_\beta + 4$. Choose a cube $Q_\beta = z_\beta + [-D_\beta^{\mathrm{buf}}/2, D_\beta^{\mathrm{buf}}/2]^3$ containing $\operatorname{supp}\beta + [-2,2]^3$. Use the centered differences $\delta_{\eta,\mu}$ and cube average J_η defined in the introduction. Put

$$a'_{\eta,\nu} = -\sum_{\mu\neq\nu} \delta_{\eta,\mu}\beta_{\mu\nu},$$
$$f^\eta_{\mu\nu} = \delta_{\eta,\mu}a'_{\eta,\nu} - \delta_{\eta,\nu}a'_{\eta,\mu}. \tag{7}$$

All these functions are smooth and real. Since each centred difference enlarges support by at most $\eta/2$ in its direction,

$$\operatorname{supp} f^\eta,\ \operatorname{supp} J_\eta a'_\eta \subset \operatorname{supp}\beta + [-\eta,\eta]^3.$$

In particular their supports lie in Q_β for $0<\eta\le 1$.

On the physical lattice $\eta\mathbb{Z}^3$ set

$$b_{\eta,\mu\nu}(c) = (J_\eta\beta_{\mu\nu})(c+\eta(e_\mu+e_\nu)/2), \qquad a^o_\eta = d^*_\eta b_\eta.$$

Here d_η is the mesh-normalized forward cochain differential and d^*_η its adjoint for the pairing $\eta^3\sum$. The adjoint formula gives

$$a^o_{\eta,\nu}(c) = (J_\eta a'_{\eta,\nu})(c+\eta e_\nu/2). \tag{8}$$

Transport to the unit lattice and define

$$\hat{\mathfrak{a}}^o_\eta(x,x+e_\nu) = \eta^{1/2}a^o_{\eta,\nu}(\eta x).$$

Its coboundary uses unscaled unit-lattice differences. The midpoint identities yield

$$d\hat{\mathfrak{a}}^o_\eta(\sigma) = f^\eta_{[\eta]}(\sigma). \tag{9}$$

Indeed, the bond values in the curl are evaluated at the two points displaced by $\pm\eta e_\mu/2$ from the plaquette centre. Their difference is $\eta\delta_{\eta,\mu}J_\eta a'_{\eta,\nu}$. Commutation of the average and differences then gives $\eta^{3/2}J_\eta f^\eta$ at that centre, which is the right side of (9).

For a compactly supported smooth two-form g, let $g_{[\eta]}$ be the covering plaquette cochain with value $\eta^{3/2}J_\eta g$ at the plaquette center. On a torus $C(n)$ of physical side $R=n\eta$, periodize this cochain by summing its translates by $n\mathbb{Z}^3$. Write $F^\sharp$ for the periodic componentwise step field whose value on the physical cube centered at a plaquette is $\eta^{-3/2}F^{\mathrm{u}}$. Then $F^\sharp(g)$ is the finite-torus pairing of the periodized $g_{[\eta]}$ with F^{u}. Periodizing the source likewise gives the exact observable identity

$$\langle d\hat{\mathfrak{a}}_\eta, F^{\mathrm{u}}\rangle = F^\sharp(f^\eta). \tag{10}$$

The source is transverse: $a^o_\eta = d^*_\eta b_\eta$ is orthogonal to every closed bond cochain, by summation by parts. This applies to both the covering cochain and its periodization.

The continuum approximation has the uniform bound

$$\|f^\eta - f\|_{H^s} \le \eta^2\|\beta\|_{H^{s+4}} \qquad (s\in\mathbb{R},\ 0<\eta\le 1). \tag{11}$$

To see this, the difference symbol is $is_\mu(p)$ with $s_\mu(p) = 2\eta^{-1}\sin(\eta p_\mu/2)$. One has $|s_\mu|\le|p_\mu|$ and $|s_\mu - p_\mu| \le \eta^2|p_\mu|^3/24$, so

$$|s_\mu s_\nu - p_\mu p_\nu| \le \eta^2|p|^4/12.$$

Each component of $f^\eta - f$ contains four such terms. Cauchy–Schwarz on these terms, followed by summation over the three two-form components, gives $|\widehat{f^\eta - f}|^2 \le (\eta^2/3)^2|p|^8|\hat\beta|^2$. Plancherel's theorem proves (11).

2.2. **Reference geometry and placement.** Let $T^* \geq 1$ be the reference exponent furnished by the construction, and put

$$R^* = hL^{T^*}, \qquad n_* = R^*/\eta, \qquad p_{\mathrm{cell}} = n_*/L, \qquad w = p_{\mathrm{cell}}/2, \qquad \ell_c = \eta w = R^*/(2L).$$

Since h and η^{-1} are powers of the odd integer L, p_{cell} is an odd integer, $w \in \mathbb{Z} + 1/2$, and the open unit-lattice reference cells are $Q_c^\circ = w(c + (0,1)^3)$. The reference torus $C(n_*)$ has $2L$ cells per side and $N_* = 8L^3$ in all.

The probe fits the reference cells if

$$D_\beta^{\mathrm{buf}} + 3\eta \leq \ell_c. \tag{12}$$

A placement is a vector $v \in \eta\mathbb{Z}^3$ satisfying

$$|v + z_\beta - (\ell_c/2)(1,1,1)|_\infty \leq \eta/2.$$

Such a vector always exists by coordinatewise rounding. It translates Q_β into $[\eta, \ell_c - \eta]^3$: the displacement of its centre is at most $\eta/2$, and the half-side is at most $(\ell_c - 3\eta)/2$. Hence

$$\mathrm{supp}(\tau_v\beta) \subset [2 + \eta, \ell_c - 2 - \eta]^3. \tag{13}$$

Form the source $\hat{\mathfrak{a}}^o$ with this translated probe.

Every endpoint of a bond in the source support is within $3\eta/2$ in sup distance of $\mathrm{supp}(\tau_v\beta)$. By (13), its lattice coordinates therefore lie in $[2/\eta - 1/2, w - 2/\eta + 1/2]$. If a plaquette has nonzero $d\hat{\mathfrak{a}}^o$, at least one of its boundary bonds belongs to that support. Its other two corners are one unit away. Thus the whole plaquette lies in the box

$$B_0 = (w/2)(1,1,1) + [-r_0, r_0]^3, \qquad r_0 = w/2 - 2/\eta + 3/2. \tag{14}$$

Condition (12) gives $r_0 \geq 3$. The distance of this box from each bounding wall is $2/\eta - 3/2 \geq 1/2$, so $B_0 \subset Q_0^\circ$. Its reflected copies lie strictly inside the corresponding cells. This is the support margin needed by the local source identity.

2.3. **Orthogonal sources and translated observables.** For a cell index c, define the wall map g_c coordinatewise: if $c_\mu = 2i$, it sends x_μ to $x_\mu + 2iw$; if $c_\mu = 2i+1$, it sends x_μ to $(2i+2)w - x_\mu$. It carries Q_0° to Q_c° and acts on cochains by transporting oriented bonds, with a sign upon orientation reversal. Put $\hat{\mathfrak{a}}_c^o = (g_c)_*\hat{\mathfrak{a}}^o$ on the covering lattice and periodize it on $C(n)$, where p_{cell} divides n and $n \geq n_*$, to obtain $\hat{\mathfrak{a}}_c$. The period lattice identifies cell indices modulo $(2n/p_{\mathrm{cell}})\mathbb{Z}^3$. The coboundary commutes with these actions and periodization, and its support lies in $B_c = g_c B_0$. For cell representatives on this torus,

$$\langle d\hat{\mathfrak{a}}_c, d\hat{\mathfrak{a}}_{c'}\rangle_n = v_\eta\, \mathbf{1}_{\{c=c'\}}, \qquad v_\eta = \|d\hat{\mathfrak{a}}^o\|^2. \tag{15}$$

The covering norm is the unweighted ℓ^2 norm, and the torus pairing is the counting pairing.

To prove the identity, unfold one factor:

$$\langle d\hat{\mathfrak{a}}_c, d\hat{\mathfrak{a}}_{c'}\rangle_n = \sum_\sigma \sum_{z\in\mathbb{Z}^3} d\hat{\mathfrak{a}}_c^o(\sigma) d\hat{\mathfrak{a}}_{c'}^o(\sigma + nz).$$

A nonzero summand has $\sigma \subset Q_c^\circ$ and $\sigma + nz \subset Q_{c'}^\circ$. The open cells are disjoint, so necessarily $c' = c + (2n/p_{\mathrm{cell}})z$. For distinct torus cells there is no such z. For the same cell exactly one translate can contribute; its summands are the squares of $d\hat{\mathfrak{a}}_c^o$. The wall action preserves the counting norm, which proves (15). Changing the placement by a vector in $\eta\mathbb{Z}^3$ translates the covering source by an integer vector. Thus v_η depends only on β and η, and is independent of the torus and the placement.

For a closed cochain u define $X_c(u) = \langle d\hat{\mathfrak{a}}_c, u\rangle_n$. Every polynomial in this observable is read by one and the same local integrand on all these tori. On the cover its local expression is

$$\widehat{A} \longmapsto P\left(\sum_{\sigma\subset B_c} d\hat{\mathfrak{a}}_c^o(\sigma)(d\widehat{A})(\sigma)\right).$$

This is a scalar gauge-invariant polynomial. Cauchy–Schwarz bounds its argument by

$$\sqrt{v_\eta}\left(\sum_{\sigma\subset B_c}|d\widehat{A}(\sigma)|^2\right)^{1/2}.$$

The polynomial is therefore bounded by

$$C_{\mathrm{pol}}\exp\left(\frac{1}{4}\sum_{\sigma\subset B_c}|d\widehat{A}(\sigma)|^2\right)$$

for a finite constant C_{pol} depending on the polynomial and the covering source. This is the local Gaussian-growth condition used for the observable class $\mathcal{A}(Q_c^\circ)$ in [26e, §4.1, (46)].

The local boxes embed in the torus. Indeed, writing $p_{\mathrm{cell}} = 2m_{\mathrm{cell}} + 1$, condition (12) gives $m_{\mathrm{cell}} \geq 7$, and

$$2r_0 + 2 = w - 4/\eta + 5 \leq w + 1 < p_{\mathrm{cell}} \leq n.$$

For each c, the translates of $B_c + [-1,1]^3$ by nz, $z \in \mathbb{Z}^3$, are consequently disjoint. On a local lift, $d\widehat{A} = F^{\mathrm{u}}$; unfolding the finite cochain sum reads the polynomial as $P(X_c)$. For a real polynomial its reflected copy from the base cell is exactly $P(X_c)$, since conjugation fixes its coefficients. This applies to the centered Wick polynomials.

The variance in (15) approaches the continuum normal-ordering variance:

$$\left|\sqrt{v_\eta} - \|f\|_2\right| \leq 21\eta|\beta|_{C^3}(D_\beta^{\mathrm{buf}})^{3/2} + \eta^2\|\beta\|_{H^4}. \tag{16}$$

Let $\Pi_\eta f^\eta$ be the componentwise cell-average step function on a period cube. By (9), its L^2 norm is $\sqrt{v_\eta}$. Each first derivative of $f^\eta_{\mu\nu}$ is bounded by $4|\beta|_{C^3}$, from the four centred-difference terms in (7) and the identity $\delta_{\eta,\mu}g = \int_{-1/2}^{1/2}(\partial_\mu g)(\,\cdot + t\eta e_\mu)\,\mathrm{d}t$. The pointwise error of the cell average is at most $12\eta|\beta|_{C^3}$. It vanishes outside the periodic images of the buffered cube, whose measure in one period is $(D_\beta^{\mathrm{buf}})^3$. Summing the three component squares bounds the L^2 error by $21\eta|\beta|_{C^3}(D_\beta^{\mathrm{buf}})^{3/2}$. The norm triangle inequality and (11) prove (16).

Finally the even cell indices give pure translations. For $y \in (R^*/L)\mathbb{Z}^3 = p_{\mathrm{cell}}\eta\mathbb{Z}^3$ set $c(y) = 2y/(p_{\mathrm{cell}}\eta) \in 2\mathbb{Z}^3$. Then $g_{c(y)}$ is the unit-lattice translation by y/η, and

$$X_{c(y)} = F^\sharp(\tau_{v+y}f^\eta). \tag{17}$$

Two such cells agree on $C(n)$ exactly when their physical translations agree modulo $R\mathbb{Z}^3$. Any fixed finite set of distinct translations therefore occupies distinct cells once the volume is sufficiently large. Lattice translation invariance removes the common placement v from the expectation of any polynomial in these observables. Its integrability follows from Gaussian domination. These identities enter the subsequent volume transfer of [26k, Theorem A], which supplies the larger-volume estimates for the translates $\tau_y f^\eta$ and their passage to component cumulants.

3. Exact external-response identities

We prove the exact finite-volume identities of [26i, §20]. We express the generating function of a curvature observable as a determinant in an external connection. The identity is first proved in each integer flux sector. In the zero-flux sector an unfolding of the integer cochains replaces the compact photon integral by a Gaussian integral over transverse cochains and an average over the three twists. We compute its normalization before applying the source shift.

3.1. **The determinant at complex connections.** Use lattice units and fix $q > 0$, $r_q = 2\pi/q$ and a real lattice mass μ_{lat}. Put $M_{\mathrm{ch}} = \mu_{\mathrm{lat}} + \frac{\rho}{2}\sum_i w_i$. For the law in theorem A, $\mu_{\mathrm{lat}} = \boldsymbol{\mu}_N$. The weights are the standard chamber weights, and $0 < \rho < 1$. For an integer triple $n = (n_1, n_2, n_3)$ let

$$V = |n| = n_1 n_2 n_3, \qquad B_n = \prod_{\mu=1}^{3} \{0, \dots, n_\mu - 1\}, \qquad \mathcal{L}_n = B_n \times \{1, 2, 3\}.$$

In this subsection each n_μ may be as small as one. Cochains are periodic functions on the oriented covering cells, so the $3V$ link orbits remain distinct even at those small sides. The chamber spinors are antiperiodic in each direction and have dimension $192V$. Write D_B for the massless chamber operator with phases $\exp(iqB)$, and $\det_n$ for its finite-dimensional determinant. In the phase convention of section 1.1, $D_B = D^{\mathrm{ch}}[qB]$. The weight $1/48$ in the spinor pairing does not rescale this determinant.

The determinant is a polynomial in the $6V$ variables $\exp(iqB(b))$ and $\exp(-iqB(b))$, of degree at most 32 in each. In particular it is entire in the complex link values. To see the degree bound, choose the site basis on $B_n \times O_{\mathrm{cub}}$ and include the antiperiodic sign whenever a neighbor is reduced into B_n. Each of the four rows at a site (c, g) has constant entries except for the one external-hop block. If $ge_1 = e_\mu$, this block contains $\exp(iqB_\mu(c))$; if $ge_1 = -e_\mu$, it contains $\exp(-iqB_\mu(c - e_\mu))$. For a fixed oriented photon link there are eight labels at each end. Thus each phase variable occurs in exactly $8 \cdot 4 = 32$ rows. The Leibniz determinant chooses one entry in each row, proving the bound. The five site labels of the diagonal and the four hops are distinct, even if a side is one, so no other phase is hidden in one of these blocks.

The two invariances needed below hold for complex cochains:

$$\det_n(D_{B+r_q k} + \mu_{\mathrm{lat}}) = \det_n(D_B + \mu_{\mathrm{lat}}), \qquad \det_n(D_{B+d\lambda} + \mu_{\mathrm{lat}}) = \det_n(D_B + \mu_{\mathrm{lat}}), \tag{18}$$

where k is integer valued and λ is a periodic complex zero-cochain. The first identity follows directly from the phases. For the second, multiplication by $\exp(iq\lambda(c))$ preserves antiperiodicity and has inverse multiplication by $\exp(-iq\lambda(c))$. On an external hop the factor

$$\mathrm{e}^{iq(B-d\lambda)(c,c')}\mathrm{e}^{iq\lambda(c')} = \mathrm{e}^{iq\lambda(c)}\mathrm{e}^{iqB(c,c')}$$

gives $D_{B-d\lambda} = \mathrm{e}^{iq\lambda} D_B \mathrm{e}^{-iq\lambda}$. Internal hops lie over one photon site and give the same conjugation. This proves the determinant identity.

For $s \geq 0$ define

$$K_n(s) = \left(|M_{\mathrm{ch}}| + \frac{1+\rho}{2}\left(\sum_i w_i - 6 + 6\mathrm{e}^{qs} \right) \right)^{192V},$$
$$|\det_n(D_B + \mu_{\mathrm{lat}})| \leq K_n(\|\operatorname{Im} B\|_\infty). \tag{19}$$

Indeed each chamber bond involution permutes the site representatives. For an internal hop its phase has modulus one; for an external hop the modulus is at most $\mathrm{e}^{q\|\operatorname{Im} B\|_\infty}$. The hopping matrix has norm $w_i(1+\rho)/2$. The operator norm is therefore bounded by the base of the power in (19), and a determinant in dimension $192V$ is bounded by that norm to the same power. This estimate is uniform in the real part of the connection.

The twist connection is

$$H_\theta = \sum_\mu \frac{\theta_\mu}{q n_\mu} \mathbf{1}_\mu, \qquad \mathbf{1}_\mu = \sum_{c \in B_n} \mathbf{1}_{(c,\mu)}. \tag{20}$$

Its determinant has period 2π in each complex twist variable. For later use we need the cochain implementing this period. Set

$$k_\nu^{(\mu)}(c) = \delta_{\mu\nu} 1_{\{c_\mu = n_\mu - 1\}}, \qquad \lambda^{(\mu)}(c) = -\frac{r_q}{n_\mu} c_\mu, \qquad c \in B_n,$$

with periodic extension. On links not crossing the μ seam, $d\lambda^{(\mu)} = -r_q/n_\mu$ in direction μ and zero in the other directions. On a seam link its value is $r_q - r_q/n_\mu$. Thus

$$H_{2\pi e_\mu} = r_q k^{(\mu)} - d\lambda^{(\mu)}. \tag{21}$$

The two invariances in (18) prove the asserted twist period. For $n_\mu = 1$, $\lambda^{(\mu)} = 0$ and $k^{(\mu)} = \mathbf{1}_\mu$, so the same formula holds.

3.2. **Real and integer cochains on the torus.** From now on $n_\mu \geq 4$. The counting pairings on cochains are extended bilinearly when their arguments are complex. Use

$$V_{\mathrm{g}} = dC^0(n), \qquad V_{\mathrm{h}} = \mathrm{span}_{\mathbb{R}}\{\mathbf{1}_1, \mathbf{1}_2, \mathbf{1}_3\}, \qquad V_{\mathrm{t}} = (V_{\mathrm{g}} + V_{\mathrm{h}})^{\perp}.$$

We recall the cochain arguments needed for the unfolding and for its Jacobian. In particular the integer condition in the flux sum requires an integer primitive, not merely a real one.

Let

$$\mathcal{P}_n^{\mathrm{per}} = \{p : p_\mu = 2\pi j_\mu/n_\mu,\ 0 \leq j_\mu < n_\mu\}, \qquad e_p(c) = V^{-1/2}\mathrm{e}^{ip\cdot c}, \qquad \xi_\mu(p) = \mathrm{e}^{ip_\mu} - 1.$$

The characters e_p are orthonormal. At each momentum p the coboundary on exterior-valued cochains is exterior multiplication by $\xi(p)$. If $\iota_{\bar\xi}$ denotes contraction by $\bar\xi$, then

$$\iota_{\bar\xi}(\xi \wedge u) + \xi \wedge \iota_{\bar\xi} u = |\xi|^2 u.$$

For $p \neq 0$ a closed coefficient u is consequently $\xi \wedge (|\xi|^{-2}\iota_{\bar\xi} u)$, and is exact. At zero momentum the cochains are constant and the coboundary vanishes. Nonzero and zero modes are orthogonal, and taking real parts of a complex primitive gives a real primitive for a real cochain. Therefore

$$\ker(d : C^1 \to C^2) = V_{\mathrm{g}} \oplus V_{\mathrm{h}}, \qquad C^1 = V_{\mathrm{g}} \oplus V_{\mathrm{h}} \oplus V_{\mathrm{t}}, \qquad \ker(d : C^2 \to C^3) = dC^1 \oplus \mathrm{Harm}^2. \tag{22}$$

Here Harm^2 is the space of constant two-cochains. The same Fourier formula gives $d^2 = 0$.

For a closed integer two-cochain z, define $w_{\mu\nu}(z)$ as its sum over one coordinate (μ, ν) two-torus. The sum is independent of the transverse coordinate: sum $dz = 0$ over the layer of cubes between two consecutive planes. Differences in the two tangential directions cancel by periodicity; the remaining difference is that of the two plane sums. The fluxes give

$$z \in dC^1(n;\mathbb{Z}) \quad \Longleftrightarrow \quad w_{\mu\nu}(z) = 0 \text{ for all } \mu < \nu. \tag{23}$$

The integer coefficients follow from the cellular cohomology of the product of the three subdivided circles. Its cochain complex is precisely the integer complex used here. The three coordinate two-tori give a basis of $H_2(T^3;\mathbb{Z})$, and $H_1(T^3;\mathbb{Z})$ is free. The universal coefficient theorem identifies $H^2(T^3;\mathbb{Z})$ with $\mathrm{Hom}(H_2(T^3;\mathbb{Z}),\mathbb{Z})$ by evaluation. The class of z evaluates to $w_{\mu\nu}(z)$ on each basis two-torus, so it vanishes exactly when these sums vanish. Vanishing of this cellular class is membership in $dC^1(n;\mathbb{Z})$. [Hat02, Theorems 3.2, 3.5 and 3B.6]

We next describe the closed integer one-cochains. If $d\lambda$ is integer valued, its sum along a path from 0 to c shows that $\lambda(c) - \lambda(0)$ is integer valued. If $d\lambda = 0$, the same path argument shows that λ is constant. It follows that

$$C^1(n;\mathbb{Z}) \cap V_{\mathrm{g}} = dC^0(n;\mathbb{Z}).$$

Each seam cochain $k^{(\mu)}$ in (21) is closed, because its multiple by r_q lies in $V_{\mathrm{g}} + V_{\mathrm{h}}$. For an arbitrary closed integer one-cochain k, (22) gives $k = d\lambda + \sum_\mu t_\mu \mathbf{1}_\mu$. Its sum along a periodic μ loop is $n_\mu t_\mu$, an integer j_μ. Subtracting $\sum_\mu j_\mu k^{(\mu)}$ leaves an integer exact cochain, so

$$k = d\lambda' + \sum_\mu j_\mu k^{(\mu)}, \qquad \lambda' \in C^0(n;\mathbb{Z}), \quad j \in \mathbb{Z}^3. \tag{24}$$

The harmonic projection of k determines j, and then determines $d\lambda'$. Moreover

$$r_q k = \left(r_q d\lambda' + \sum_\mu j_\mu d\lambda^{(\mu)} \right) + H_{2\pi j}. \tag{25}$$

The first term is exact and the second is harmonic. Their coupling through j will matter when choosing a fundamental domain.

The scalar Laplacian $\Delta_0 = d^*d$ has eigenvalues $|\xi(p)|^2 = \sum_\mu (2 - 2\cos p_\mu)$ and a one-dimensional kernel. Write

$$\det{}'\Delta_0 = \prod_{p \in \mathcal{P}_n^{\mathrm{per}} \setminus \{0\}} |\xi(p)|^2, \qquad \mathcal{L}(n) = \log \det{}'\Delta_0. \tag{26}$$

At each nonzero momentum the transverse one-cochains are the two-dimensional space $\xi(p)^\perp$. For $v \in \mathbb{C}^3$,

$$\sum_{\mu<\nu} |\xi_\mu v_\nu - \xi_\nu v_\mu|^2 = |\xi|^2 |v|^2 - |\bar{\xi} \cdot v|^2.$$

Thus d^*d acts on that transverse space as multiplication by $|\xi(p)|^2$, and

$$\dim V_{\mathrm{t}} = 2V - 2, \qquad \det(d^*d|_{V_{\mathrm{t}}}) = (\det{}'\Delta_0)^2. \tag{27}$$

Diagonalizing this positive real form gives its Gaussian integral

$$\mathcal{G}_n := \int_{V_{\mathrm{t}}} \mathrm{e}^{-\|dA\|^2/2} \, \mathrm{d}\sigma(A) = \frac{(2\pi)^{V-1}}{\det{}'\Delta_0}. \tag{28}$$

The measure σ is Euclidean Lebesgue measure for the counting pairing, not a coordinate measure chosen along a nonorthogonal basis.

There is one more determinant needed for the exact directions. The cochains $d\mathbf{1}_c$, $c \in B_n \setminus \{0\}$, form both a real basis of V_{g} and an integer basis of $dC^0(n;\mathbb{Z})$. Indeed one subtracts the value at zero from a zero-cochain; independence follows because the resulting zero-cochain has value zero there and cannot be a nonzero constant. Their Gram determinant is

$$\det\left(\langle d\mathbf{1}_c, d\mathbf{1}_{c'} \rangle \right)_{c,c' \neq 0} = \frac{\det{}'\Delta_0}{V}. \tag{29}$$

To prove this formula, let S be the scalar Laplacian matrix in the site basis. The coefficient of t in $\det(t + S)$ is $\det{}'\Delta_0$, by its eigenvalues. Multilinearity of the determinant in columns identifies that coefficient with the sum of the V principal minors obtained by deleting one row and column. All these minors have the same determinant, since site translations conjugate S by permutations. The minor at zero is the Gram matrix above.

3.3. **Unfolding the zero-flux sector.** Let Φ be a Borel function on $C^2(n)$ with $|\Phi(u)| \le C_\Phi \mathrm{e}^{C_\Phi \|u\|}$. Let W be a bounded continuous complex function on $C^1(n)$, invariant under addition of real exact cochains and of r_q times integer one-cochains. Then

$$\begin{aligned} &\sum_{\substack{z \in C^2(n;\mathbb{Z}),\, dz=0 \\ w(z)=0}} \int_{[0,r_q)^{\mathcal{L}_n}} \Phi(dA + r_q z) \mathrm{e}^{-\|dA + r_q z\|^2/2} W(A) \, \mathrm{d}A \\ &\quad = \mathcal{N}_n \int_{[-\pi,\pi]^3} \frac{\mathrm{d}\theta}{(2\pi)^3} \int_{V_{\mathrm{t}}} \Phi(dA) \mathrm{e}^{-\|dA\|^2/2} W(A + H_\theta) \, \mathrm{d}\sigma(A), \end{aligned} \tag{30}$$

$$\mathcal{N}_n = r_q^{V+2} (\det{}'\Delta_0)^{1/2}. \tag{31}$$

Both sides converge absolutely. The coefficient comes from the lattice of exact gauge transformations and the three periods of the harmonic directions; we give the unfolding with both contributions.

First, set

$$C_q(\epsilon) = \sup_{y \in \mathbb{R}} \sum_{j \in \mathbb{Z}} \mathrm{e}^{-\epsilon (y + r_q j)^2} \le 2 + r_q^{-1} \sqrt{\pi/\epsilon} \qquad (\epsilon > 0).$$

The bound follows by keeping the two nearest terms and comparing the remaining terms with Gaussian integrals over adjacent intervals, as in [26e, (5)]. Summing each integer coordinate gives

$$\sum_{z\in C^2(n;\mathbb{Z})}\int_{[0,r_q)^{\mathcal{L}_n}} \mathrm{e}^{-\epsilon\|dA+r_qz\|^2}\,\mathrm{d}A \le (r_qC_q(\epsilon))^{3V}.$$

Here the integer closedness and flux restrictions have been dropped. Completing the scalar square bounds $|\Phi(u)|\mathrm{e}^{-\|u\|^2/2}$ by $C_\Phi \mathrm{e}^{C_\Phi^2}\mathrm{e}^{-\|u\|^2/4}$. Together with boundedness of W, this proves absolute convergence of the left side.

Let $\mathcal{Z}^1_{\mathbb{Z}} = \ker d \cap C^1(n;\mathbb{Z})$ and choose representatives $\mathcal{R}$ of $C^1(n;\mathbb{Z})/\mathcal{Z}^1_{\mathbb{Z}}$. By (23), $k \mapsto dk$ is a bijection from $\mathcal{R}$ onto the closed zero-flux integer two-cochains. Put

$$P(A) = \Phi(dA)\mathrm{e}^{-\|dA\|^2/2}W(A), \qquad \mathcal{D}_1 = \bigcup_{k\in\mathcal{R}}([0,r_q)^{\mathcal{L}_n} + r_qk).$$

The two invariances of W turn the left side of (30) into $\int_{\mathcal{D}_1} P(A)\,\mathrm{d}A$. The boxes are disjoint. Every A has a unique decomposition into its coordinatewise remainder in $[0,r_q)$ and r_q times an integer cochain. Decomposing that cochain into its chosen representative and a closed one shows that $\mathcal{D}_1$ is a Borel fundamental domain for $r_q\mathcal{Z}^1_{\mathbb{Z}}$. The function P is invariant under that group.

We replace this domain by

$$\mathcal{F}_{\mathrm{g}} = \left\{ r_q\sum_{c\neq 0} t_c\, d\mathbf{1}_c : 0 \le t_c < 1 \right\}, \qquad \mathcal{F}_{\mathrm{h}} = \{H_\theta : \theta \in [-\pi,\pi)^3\}, \qquad \mathcal{D}_2 = \mathcal{F}_{\mathrm{g}} + \mathcal{F}_{\mathrm{h}} + V_{\mathrm{t}}.$$

To check that it is a fundamental domain for the same group, write $A = A_{\mathrm{g}} + A_{\mathrm{h}} + A_{\mathrm{t}}$. Formula (25) shows that subtracting r_qk changes the harmonic part by $H_{2\pi j}$ and the exact part by $\sum_\mu j_\mu d\lambda^{(\mu)} + r_q d\lambda'$. There is exactly one j placing the harmonic part in $\mathcal{F}_{\mathrm{h}}$. For this j there is exactly one $d\lambda'$ placing the exact part in $\mathcal{F}_{\mathrm{g}}$, by the integer basis used in (29). The transverse part is unchanged. Uniqueness in (24) gives uniqueness of k. The orthogonal decomposition identifies $\mathcal{D}_2$ with a product of Borel sets, so it is Borel.

For an integrable periodic Borel function, the integral over two such fundamental domains is the same. Indeed partition $\mathcal{D}_1$ into its intersections with $\mathcal{D}_2 + h$, $h \in r_q\mathcal{Z}^1_{\mathbb{Z}}$, and translate each intersection by $-h$. They partition $\mathcal{D}_2$, and invariance of both the measure and the function gives equality. Tonelli proves it first for nonnegative functions; applying it to the modulus then justifies the complex case by absolute integrability. Thus $\int_{\mathcal{D}_1} P = \int_{\mathcal{D}_2} P$.

The two finite-dimensional volumes are

$$\mathrm{vol}(\mathcal{F}_{\mathrm{g}}) = r_q^{V-1}(\det{}'\Delta_0/V)^{1/2}, \tag{32}$$

$$\mathrm{d}A_{\mathrm{h}} = V^{1/2}q^{-3}\,\mathrm{d}\theta = r_q^3V^{1/2}\frac{\mathrm{d}\theta}{(2\pi)^3}. \tag{33}$$

The first follows from the square root of the Gram determinant. For the second, the constant cochains $V^{-1/2}\mathbf{1}_\mu$ are orthonormal, and the matrix of $\theta \mapsto H_\theta$ in this basis has diagonal entries $V^{1/2}/(qn_\mu)$. Their product is $V^{1/2}q^{-3}$. Their combined factor is precisely $\mathcal{N}_n$. Since $P(A_{\mathrm{g}} + A_{\mathrm{h}} + A_{\mathrm{t}}) = \Phi(dA_{\mathrm{t}})\mathrm{e}^{-\|dA_{\mathrm{t}}\|^2/2}W(A_{\mathrm{t}} + A_{\mathrm{h}})$, it is independent of A_{g}. Fubini over the orthogonal decomposition and the two displayed volume formulas prove (30), including absolute convergence of its right side.

Apply this identity to $W(A) = \det_n(D_A + \mu_{\mathrm{lat}})$. Its hypotheses follow from (18) and (19). If $Z_n^{(0)}[\Phi]$ is the zero-flux part of the configuration integral, we have proved

$$Z_n^{(0)}[\Phi] = \mathcal{N}_n \int \frac{\mathrm{d}\theta}{(2\pi)^3}\int_{V_{\mathrm{t}}} \Phi(dA)\mathrm{e}^{-\|dA\|^2/2}\det_n(D_{A+H_\theta} + \mu_{\mathrm{lat}})\,\mathrm{d}\sigma(A). \tag{34}$$

The twist integral always runs over $[-\pi,\pi]^3$. For real A and θ the determinant is nonnegative. Also $Z_n^{(0)}[1] > 0$ at every real μ_{lat}: the lower bound in [26e, Proposition 2.3] was obtained from the single

term $z = 0$, which belongs to this sector. Dividing (34) by its value at $\Phi = 1$ identifies the zero-flux curvature law with the law of dA under the harmonic-averaged transverse measure.

3.4. **The Gaussian normalization and matter density.** Let $F_n = dC^1(n) = dV_{\mathrm{t}}$ and let γ_n be the standard Gaussian probability measure on this Euclidean space. The map $d_{\mathrm{t}} = d|_{V_{\mathrm{t}}}$ is a linear bijection onto F_n, by (22); write $A(u) = d_{\mathrm{t}}^{-1}u$. Changing variables along this map gives

$$\int_{V_{\mathrm{t}}} G(dA)\mathrm{e}^{-\|dA\|^2/2}\,\mathrm{d}\sigma(A) = \mathcal{G}_n \int G(u)\,\mathrm{d}\gamma_n(u). \tag{35}$$

Indeed the Jacobian is $|\det d_{\mathrm{t}}|^{-1}$, and $\dim F_n = 2V - 2$. The Gaussian density contributes $(2\pi)^{V-1}$; setting $G = 1$ identifies the constant with (28). The formula holds for nonnegative Borel functions, and for the exponentially bounded complex functions used above by their Gaussian integrability.

Define the photon factor and normalized matter factor by

$$Z_n^{\mathrm{ph}} = \mathcal{N}_n\mathcal{G}_n = r_q^{V+2}(2\pi)^{V-1}\mathrm{e}^{-\mathcal{L}(n)/2}, \tag{36}$$

$$z_{r,n}(q,\mu_{\mathrm{lat}}) = \int \frac{\mathrm{d}\theta}{(2\pi)^3}\int \det_n(D_{A(u)+H_\theta} + \mu_{\mathrm{lat}})\,\mathrm{d}\gamma_n(u). \tag{37}$$

Then (34) and (35) imply

$$Z_n^{(0)}[1] = Z_n^{\mathrm{ph}} z_{r,n}, \qquad 0 < z_{r,n} \le K_n(0). \tag{38}$$

Positivity follows from that of $Z_n^{(0)}[1]$; the upper bound follows because both integration measures in (37) are probability measures.

On a cubic torus $V = n^3$, fix a positive reference mass $\bar{m}$, mesh η and a real pair $(m_{\mathrm{in}}, \varepsilon_{\mathrm{in}}^{\mathrm{v}})$, with $\mu_{\mathrm{lat}} = \bar{m}\eta + m_{\mathrm{in}}$. The density with external cochain a is

$$\mathrm{dens}^{\theta,a}(u) = \mathrm{e}^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3}\frac{\det_n(D_{A(u)+H_\theta+a} + \mu_{\mathrm{lat}})}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta)}. \tag{39}$$

The denominator is positive, since the real part of $D^{\mathrm{ch}}[0] + \bar{m}\eta$ is at least $\bar{m}\eta > 0$. At $a = 0$ this is a nonnegative bounded continuous density. Let $\nu^{\theta,0}$ be its measure against γ_n and let $\bar{\nu}$ be the twist average. Equations (34) and (35) give

$$\bar{\nu}(\Phi) = \frac{\mathrm{e}^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3}}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta)Z_n^{\mathrm{ph}}} Z_n^{(0)}[\Phi], \tag{40}$$

$$\bar{\nu}(1) = \frac{\mathrm{e}^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3} z_{r,n}}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta)} \in (0,\infty). \tag{41}$$

Thus $\bar{\nu}/\bar{\nu}(1)$ is exactly the zero-flux curvature law. The vacuum normalization cancels in this quotient. The quantities Z_n^{ph} and $z_{r,n}$ depend only on $(n, q, \mu_{\mathrm{lat}})$ and the fixed chamber parameters; the auxiliary split $\mu_{\mathrm{lat}} = \bar{m}\eta + m_{\mathrm{in}}$ does not change them.

3.5. **Holomorphic external partition functions.** Write $\mathfrak{F}(n)$ for the exponentially bounded Borel functions Φ used above. For $w \in \mathbb{Z}^3$ let $\mathcal{M}_w$ be the closed integer two-cochains of flux w. For a complex one-cochain a define

$$Z_n^{\mathrm{x},(w)}(a;\Phi) = \sum_{z\in\mathcal{M}_w}\int_{[0,r_q)^{\mathcal{L}_n}} \Phi(F)\mathrm{e}^{-\|F\|^2/2}\det_n(D_{A+a} + \mu_{\mathrm{lat}})\,\mathrm{d}A, \qquad F = dA + r_q z, \tag{42}$$

$$\mathbf{Z}_n^{\mathrm{x}}(a;\Phi) = \int \frac{\mathrm{d}\theta}{(2\pi)^3}\int_{V_{\mathrm{t}}} \Phi(dA)\mathrm{e}^{-\|dA\|^2/2}\det_n(D_{A+H_\theta+a} + \mu_{\mathrm{lat}})\,\mathrm{d}\sigma(A). \tag{43}$$

When $\Phi = 1$ we omit it from the notation. These are partition functions in an external connection; the curvature F in their photon weight is not shifted.

Proposition 3.1. For every $\Phi \in \mathfrak{F}(n)$ both expressions converge absolutely, locally uniformly in a, and are entire in its $3V$ complex link values. The sum over all sectors converges absolutely, locally uniformly as well. Moreover

$$Z_n^{\mathrm{x},(0)}(a;\Phi) = \mathcal{N}_n \mathbf{Z}_n^{\mathrm{x}}(a;\Phi), \qquad Z_n^{\mathrm{x},(w)}(0;\Phi) = Z_n^{(w)}[\Phi]. \tag{44}$$

In particular $\mathbf{Z}_n^{\mathrm{x}}(0) > 0$. ∎

Proof. For real A, θ the imaginary parts of $A + a$ and $A + H_\theta + a$ are both $\operatorname{Im} a$. On a polydisc $\max_b |a(b)| \le s$ the determinant is bounded by $K_n(s)$. If $|\Phi(u)| \le C_\Phi \mathrm{e}^{C_\Phi \|u\|}$, the absolute value of each integrand is consequently bounded by

$$K_n(s) C_\Phi \mathrm{e}^{C_\Phi^2} \mathrm{e}^{-\|F\|^2/4} \quad \text{or} \quad K_n(s) C_\Phi \mathrm{e}^{C_\Phi^2} \mathrm{e}^{-\|dA\|^2/4}.$$

The first majorant can be summed over every integer two-cochain and integrated over the compact box, giving $K_n(s) C_\Phi \mathrm{e}^{C_\Phi^2} (r_q C_q(1/4))^{3V}$. The second is integrable by (28): rescaling A by $\sqrt{2}$ gives

$$\int_{V_{\mathrm{t}}} \mathrm{e}^{-\|dA\|^2/4} \, \mathrm{d}\sigma(A) = 2^{V-1} \mathcal{G}_n.$$

These bounds also show local uniform absolute convergence of the sector sum.

We recall the parameter-integral argument to justify entireness and all later differentiations. Let (Y, ν) be a σ-finite measure space. Suppose $v(y, a)$ is measurable in (y, a), entire in $a \in \mathbb{C}^d$ for each y, and bounded on each polydisc by a ν-integrable $M_s(y)$. If $v_\alpha(y)$ are its Taylor coefficients at zero, integration over the product of circles of radius s gives

$$v_\alpha(y) = s^{-|\alpha|} \int_{[0,2\pi]^d} v(y, s\mathrm{e}^{i\theta}) \mathrm{e}^{-i\alpha\cdot\theta} \frac{\mathrm{d}\theta}{(2\pi)^d}, \qquad |v_\alpha(y)| \le s^{-|\alpha|} M_s(y).$$

The integral formula makes the coefficients measurable. For $\max_j |a_j| \le s$, applying the bound at radius $2s$ yields

$$\sum_\alpha \int |v_\alpha(y) a^\alpha| \, \mathrm{d}\nu(y) \le \sum_\alpha 2^{-|\alpha|} \int M_{2s}(y) \, \mathrm{d}\nu(y) = 2^d \int M_{2s}(y) \, \mathrm{d}\nu(y) < \infty.$$

Thus the Taylor series may be integrated term by term, and its integral is an entire function with coefficients $\int v_\alpha$. This also proves differentiation under the integral at zero. Translating the parameter to an arbitrary a^0 proves it there, using the majorant $M_{s+\|a^0\|_\infty}$. This argument applies to a counting measure times Lebesgue measure as well as to the transverse integral.

Apply this argument to the two displayed majorants. The dependence on a in each integrand is a finite sum of exponentials of linear forms, by the determinant polynomial. The other factors are Borel and independent of a, so the hypotheses hold also when Φ is merely Borel. This proves the asserted holomorphy. For fixed complex a, the function $W_a(A) = \det_n(D_{A+a} + \mu_{\mathrm{lat}})$ on real cochains is bounded by $K_n(\|\operatorname{Im} a\|_\infty)$, continuous, and has the two invariances required in (30). That formula gives (44); the value at zero follows from the definitions. Its positive zero value follows from $Z_n^{(0)}[1] > 0$ and $\mathcal{N}_n > 0$. □

The complex densities in (39) define finite complex measures $\nu^{\theta,a}$ and their twist average $\bar{\nu}^a$ against γ_n. Their absolute values are bounded by

$$|\mathrm{dens}^{\theta,a}(u)| \le \frac{\mathrm{e}^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3} K_n(\|\operatorname{Im} a\|_\infty)}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta)}. \tag{45}$$

This bound is independent of (θ, u). Fubini therefore writes the averaged measure as a bounded Borel density against γ_n. Applying (35) at each twist, and then (44), gives

$$\bar{\nu}^a(\Phi) = \frac{\mathrm{e}^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3}}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta)\mathcal{G}_n} \mathbf{Z}_n^{\mathrm{x}}(a;\Phi) = \frac{\mathrm{e}^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3}}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta) Z_n^{\mathrm{ph}}} Z_n^{\mathrm{x},(0)}(a;\Phi). \tag{46}$$

The exponential majorants in the proof above justify both changes of integration for $\Phi \in \mathfrak{F}(n)$. In particular, for every Borel set $E \subset F_n$, $a \mapsto \bar{\nu}^a(E)$ is entire and

$$|\bar{\nu}^a(E)| \le \frac{\mathrm{e}^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3} K_n(\|\operatorname{Im} a\|_\infty)}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta)}.$$

Indeed F_n is a closed linear subspace, so its Borel set indicator extends to a bounded Borel function on $C^2(n)$. Composition with any linear parameter map $t \mapsto a(t)$ remains entire. These statements concern the unnormalized complex measures; the positive normalization at $a = 0$ is (41).

3.6. **The source shift and Wick derivatives.**

Proposition 3.2. For every flux w and complex one-cochain a,

$$Z_n^{(w)}[\mathrm{e}^{\langle da, F\rangle}] = \mathrm{e}^{\langle da, da\rangle/2} Z_n^{\mathrm{x},(w)}(a). \tag{47}$$

Both sides, and their sums over w, converge absolutely. Consequently

$$Z_n[\mathrm{e}^{\langle da, F\rangle}] = \mathrm{e}^{\langle da, da\rangle/2} \sum_w Z_n^{\mathrm{x},(w)}(a). \tag{48}$$

The quadratic form in the prefactor is bilinear; there is no complex conjugation of da. ■

Proof. First let a be real. On real cochains define

$$P_a(A) = \sum_{z \in \mathcal{M}_w} \mathrm{e}^{-\|dA + r_q z\|^2/2 + \langle da, dA + r_q z\rangle} \det_n(D_A + \mu_{\mathrm{lat}}).$$

This is a nonnegative Borel function. Adding $r_q \mathbf{1}_b$ to A adds $d\mathbf{1}_b$ to the integer two-cochain in its curvature. This map permutes $\mathcal{M}_w$: its image is closed and has the same flux, since each exact plaquette cochain has zero plane sums. The determinant is unchanged. Hence P_a is invariant under $r_q C^1(n;\mathbb{Z})$. Both the compact box and its translate by a are fundamental domains for this group. The domain argument in section 3.3 gives

$$\int_{[0,r_q)^{\mathcal{L}_n}} P_a(A)\,\mathrm{d}A = \int_{[0,r_q)^{\mathcal{L}_n}} P_a(A + a)\,\mathrm{d}A.$$

On the right the exponent completes to

$$-\tfrac{1}{2}\|F + da\|^2 + \langle da, F + da\rangle = -\tfrac{1}{2}\|F\|^2 + \tfrac{1}{2}\|da\|^2.$$

Tonelli now identifies the two sides with (47) for real a. Their finiteness also follows from the Gaussian majorant used below.

For complex a, the right side of (47) is entire by Proposition 3.1. The left side is entire by the same parameter-integral argument. In fact, on $\max_b |a(b)| \le s$,

$$\|d \operatorname{Re} a\| \le 4s\sqrt{3V} =: c_s, \qquad \left|\mathrm{e}^{-\|F\|^2/2 + \langle da, F\rangle} \det_n(D_A + \mu_{\mathrm{lat}})\right| \le K_n(0)\mathrm{e}^{c_s^2}\mathrm{e}^{-\|F\|^2/4}.$$

This can be summed and integrated over all integer two-cochains, so it supplies an integrable polydisc majorant. Both sides agree on the real cochain space. An entire function vanishing there has all complex partial derivatives zero there: at a real point these derivatives can be taken along real increments. Repeating this for all derivatives shows that every Taylor coefficient at zero vanishes. Thus the two entire functions agree for all complex a. Finally the same Gaussian majorant permits summation over sectors; the sector sum on the right converges absolutely by Proposition 3.1. □

Let $a^{(1)}, \dots, a^{(r)}$ be real cochains and $X_j = \langle da^{(j)}, F\rangle$. Dividing (47) at $w = 0$ by its positive zero-source value yields the normalized generating identity

$$\mathbb{E}_n^{\mathrm{av}} \exp\left(\sum_j z_j X_j - \tfrac{1}{2}\sum_{i,j} z_i z_j \langle da^{(i)}, da^{(j)}\rangle\right) = \frac{\mathbf{Z}_n^{\mathrm{x}}(\sum_j z_j a^{(j)})}{\mathbf{Z}_n^{\mathrm{x}}(0)}, \qquad z \in \mathbb{C}^r. \tag{49}$$

The expectation $\mathbb{E}_n^{\mathrm{av}}$ is the zero-flux law identified in (34). The Gaussian factor on the left is determined by the counting pairing of the curvatures of the source cochains.

Use the reference cells of Section 2 on the torus n_*. Their index set is $\mathcal{C}_* = \{0,\dots,2L-1\}^3$, and $\widehat{a}_c$ is the periodization of $\hat{\mathfrak{a}}_c^o$. These are real periodic cochains. Equation (15) gives

$$\langle d\widehat{a}_c, d\widehat{a}_{c'}\rangle = v_\eta\, 1_{\{c=c'\}}. \tag{50}$$

The variance $v_\eta = \|d\hat{\mathfrak{a}}_\eta^o\|^2$ is independent of the cell and the placement. This is the cell-geometry input needed here: the external-response proof uses neither the cell width nor its offsets otherwise. Put

$$a(t) = e^{-1}\sum_{c\in\mathcal{C}_*} t_c\widehat{a}_c, \qquad \bar{Z}^{\mathrm{ext}}(t) = \int \frac{\mathrm{d}\theta}{(2\pi)^3}\int \det_{n_*}(D_{A(u)+H_\theta+a(t)}+\mu_{\mathrm{lat}})\,\mathrm{d}\gamma_{n_*}(u).$$

The reference torus has $n_* \geq 4$, and $e>0$ is the physical charge in this convention. Equation (35) gives

$$\bar{Z}^{\mathrm{ext}}(t) = \mathcal{G}_{n_*}^{-1}\mathbf{Z}^{\mathrm{x}}_{n_*}(a(t)), \qquad \mathring{\Phi}(t) := \frac{\bar{Z}^{\mathrm{ext}}(t)}{\bar{Z}^{\mathrm{ext}}(0)} = \frac{\mathbf{Z}^{\mathrm{x}}_{n_*}(a(t))}{\mathbf{Z}^{\mathrm{x}}_{n_*}(0)}. \tag{51}$$

Its zero value in the denominator is positive, and $\mathring{\Phi}$ is entire. With $X_c = \langle d\widehat{a}_c, F\rangle$, (49) becomes

$$\mathbb{E}^{\mathrm{av}}_{n_*}\exp\sum_c (z_cX_c - \tfrac{1}{2}v_\eta z_c^2) = \mathring{\Phi}(ez). \tag{52}$$

Define the Hermite polynomials with this variance by $H_j(y) = \partial_z^j \mathrm{e}^{zy - v_\eta z^2/2}|_{z=0}$. In particular

$$H_2(y) = y^2 - v_\eta, \qquad H_4(y) = y^4 - 6v_\eta y^2 + 3v_\eta^2.$$

To differentiate (52) under the expectation, on a polydisc $\max_c|z_c|\leq s$ use

$$\left|\exp\sum_c (z_cX_c - \tfrac{1}{2}v_\eta z_c^2)\right| \leq \exp\left(s|\mathcal{C}_*|\sqrt{v_\eta}\|F\| + \tfrac{1}{2}|\mathcal{C}_*|s^2v_\eta\right).$$

This is exponentially bounded in the real curvature and is integrable under the zero-flux law, by the Gaussian majorant. The parameter-integral argument therefore applies to this probability measure. The resulting products of H_2 or H_4 are also integrable: $|X_c|\leq\sqrt{v_\eta}\|F\|$ bounds them by a finite polynomial in $\|F\|$, which is bounded by a constant times $\mathrm{e}^{\|F\|}$.

For $m=2$ or 4 and a subset $S\subset\mathcal{C}_*$, differentiate in precisely those variables. The factor ez on the right supplies $e^{m|S|}$, giving

$$\mathbb{E}^{\mathrm{av}}_{n_*}\prod_{c\in S} e^{-m}H_m(X_c) = \left(\prod_{c\in S}\partial_{t_c}^m\mathring{\Phi}\right)(0).$$

Expanding the product over subsets proves, for every real c_m,

$$\mathbb{E}^{\mathrm{av}}_{n_*}\prod_{c\in\mathcal{C}_*}(e^{-m}H_m(X_c) - c_m) = \left(\prod_{c\in\mathcal{C}_*}(\partial_{t_c}^m - c_m)\mathring{\Phi}\right)(0). \tag{53}$$

The derivative operators commute because $\mathring{\Phi}$ is entire. This identity is the finite-volume quantity to which the terminal interacting estimates will be applied; all its normalization factors have already been included in (51).

4. Reference scales and rescaled probes

We use the reference torus of the tuned flow and prove the uniform probe estimates from [26i, §21.1]. Fix $\bar m \in (0,1]$, put $h = L^{\lfloor \log_L(1/\bar m)\rfloor}$, and assume $0 < e\sqrt{h} \le e_1$ and $N \ge N_{\mathrm{ref}}(e)$. By [26a, (47)] and [26a, (53)],

$$\begin{aligned} \eta = L^{-N}, \qquad K = N + \log_L h, \qquad a_j = L^{j-N}, \\ e_j = e\sqrt{a_j}, \qquad \eta_j = L^{-j} = \eta/a_j, \qquad \ell_j = R^*/a_j. \end{aligned} \tag{54}$$

Thus a point with level-j coordinates c has physical coordinates $a_j c$, and $T_j = \eta_j\mathbb{Z}^3/\ell_j\mathbb{Z}^3$. At $j=0$ the charge is $q = e\sqrt{\eta}$. The symbols e_0, e_1 without a running index retain their meanings as charge ceilings. Set $v_j = -\log e_j$, $p_{0,j} = v_j^{p_0}$ and $p_j = v_j^p$; r_j is the least nonnegative integer power of L at least $v_j^{r_{\mathrm{loc}}}$. For a bond cochain a on the fine unit torus, its rescaling is

$$a^{[j]}(b) = L^{j/2} a(L^j b). \tag{55}$$

The physical scale a_j is a number, while a denotes a connection. On the reference torus, [26a, (54)] and [26a, (50)] give

$$k_* \in \{K-1, K-2\}, \qquad a_* \in \{h/L, h/L^2\}, \qquad \ell_* = Mr_*, \qquad r_* > 31L + 16.$$

These stop-scale quantities are independent of $N \ge N_{\mathrm{ref}}(e)$. Write r_{loc} for the localization power in [26a, §6.1].

Let $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ be real and nonzero. Use the cube $Q_\beta = z_\beta + [-D_\beta^{\mathrm{buf}}/2, D_\beta^{\mathrm{buf}}/2]^3$ of Section 2, where $D_\beta^{\mathrm{buf}} = D_\beta + 4$. In this section a placement $v \in \eta\mathbb{Z}^3$ is admitted if, in physical coordinates,

$$v + Q_\beta \subset [\ell_c/8, 7\ell_c/8]^3, \qquad \ell_c = R^*/(2L). \tag{56}$$

A centered placement has $|v + z_\beta - (\ell_c/2)(1,1,1)|_\infty \le \eta/2$. It is admitted when $D_\beta^{\mathrm{buf}}/2 + \eta/2 \le 3\ell_c/8$. This follows by adding the displacement of the cube center to its half-side.

Form $\widehat{\mathfrak{a}}$ with $\tau_v\beta$ as in (8), and periodize it on $C(n_*)$, $n_* = R^*/\eta$. For $c \in \mathcal{C}_* = \{0, \dots, 2L-1\}^3$, write $\widehat{\mathfrak{a}}_c$ for its cell copy; thus $\widehat{\mathfrak{a}}_c = \widehat{a}_c$ in the introduction. In physical coordinates the wall map carrying the first cell to cell c acts in coordinate μ by translation through $2i\ell_c$ if $c_\mu = 2i$, and by reflection $y_\mu \mapsto (2i+2)\ell_c - y_\mu$ if $c_\mu = 2i+1$. Define the marked connections by

$$a(t) = e^{-1}\sum_c t_c \widehat{\mathfrak{a}}_c, \qquad \widehat{c}_c = e^{-1}\widehat{\mathfrak{a}}_c^{[k_*]}, \qquad \mathscr{C}(t) = a(t)^{[k_*]} = \sum_c t_c \widehat{c}_c. \tag{57}$$

For a level-j bond function A, ∂A uses forward differences divided by η_j, and

$$|\delta_\alpha \partial A|_\infty = \sup_{\mu,\nu} \sup_{0 < d(x,x') \le 1} \frac{|\partial_\nu A_\mu(x') - \partial_\nu A_\mu(x)|}{d(x,x')^\alpha}.$$

The external norm of a fine connection is $|a|_{\mathrm{ext}} = |a^{[k_*]}|_\infty + |\partial a^{[k_*]}|_\infty + |\delta_\alpha \partial a^{[k_*]}|_\infty$.

We define the uniform probe constants over both possible stop scales. For $\eta' = L^{-N'}$ put

$$g_\nu^{\eta'}[\beta'] = -\sum_{\mu \ne \nu} \delta_{\eta',\mu} J_{\eta'} \beta'_{\mu\nu}.$$

This is the smooth vector field $J_{\eta'} a'_{\eta'}$ in the midpoint formula. Let $\mathcal{I}_\beta$ consist of triples (N', a_{s}, r') with $N' \ge 0$, $a_{\mathrm{s}} \in \{h/L, h/L^2\}$, $N' + \log_L a_{\mathrm{s}} \ge 1$, and $r' \in \{\pm 1\}^3$. r' is identified with its diagonal orthogonal matrix, and $U_{r'}\beta(x) = (\wedge^2 r')\beta(r'^{-1}x)$. At such a triple set $\eta'' = \eta'/a_{\mathrm{s}}$, $g^0 = g^{\eta'}[U_{r'}\beta]$, and

$$G_\nu(c) = \sqrt{a_{\mathrm{s}}}\, g_\nu^0(a_{\mathrm{s}} c + \eta' e_\nu/2) \quad (c \in \eta''\mathbb{Z}^3), \qquad W_\nu(x) = \sqrt{\eta'}\, g_\nu^0(\eta' x + \eta' e_\nu/2) \quad (x \in \mathbb{Z}^3).$$

The three norms of G are taken at mesh η''. Set

$$C_\beta = \sup_{\mathcal{I}_\beta}(|G|_\infty + |\partial G|_\infty + |\delta_\alpha \partial G|_\infty), \qquad \mathbf{D} = \sup_{\mathcal{I}_\beta} \|dW\|^2_{\ell^2(\text{plaquettes})}. \tag{58}$$

These suprema depend on $\bar{m}, L, \alpha, \beta$, and contain no charge, torus, or placement. They have the finite bounds

$$C_\beta \le 2\sqrt{h}(1+h+3h^2)|\beta|_{C^3}, \qquad \mathbf{D} \le 48|\beta|_{C^2}^2(D_\beta^{\mathrm{buf}}+3)^3. \tag{59}$$

Indeed the centered difference is the average of the corresponding derivative along its segment, and $J_{\eta'}$ has norm one on each derivative sup norm. Each component of g^0 has two summands, hence $|D^i g^0_\nu|_\infty \le 2|\beta|_{C^{i+1}}$ for $i = 0, 1, 2$. This gives $|G|_\infty \le 2a_{\mathrm{s}}^{1/2}|\beta|_{C^1}$. A first lattice difference of G is $a_{\mathrm{s}}^{3/2}$ times an average of $\partial_\mu g^0_\nu$, so $|\partial G|_\infty \le 2a_{\mathrm{s}}^{3/2}|\beta|_{C^2}$. The difference of these averages at c, c' is bounded by $6a_{\mathrm{s}}|c-c'|\,|\beta|_{C^3}$. Since $|c-c'| \le 1$ and $\alpha < 1$, the third norm is at most $6a_{\mathrm{s}}^{5/2}|\beta|_{C^3}$. Sum these bounds and use $a_{\mathrm{s}} \le h$.

For the energy, each curl component of W is a difference of two first differences, and has modulus at most $4(\eta')^{3/2}|\beta|_{C^2}$. A supporting plaquette has a supporting bond, whose endpoints in physical coordinates lie in $r'Q_\beta$. Its basepoint is therefore in that cube enlarged by η' in each negative coordinate direction. There are at most $(D_\beta^{\mathrm{buf}}/\eta' + 2)^3$ basepoints and three plaquette directions. Squaring the pointwise curl bound gives $48|\beta|_{C^2}^2(D_\beta^{\mathrm{buf}} + 2\eta')^3$, which implies (59).

At the reference stop, $(N, a_*, 1) \in \mathcal{I}_\beta$. The midpoint formula and (55) give

$$\widehat{\mathfrak{a}}_\nu^{[k_*]}(c) = \sum_{z\in\mathbb{Z}^3} G_\nu(c - v/a_* + \ell_* z).$$

Supporting bond endpoints lie in $a_*^{-1}(v + Q_\beta)$. Distinct periodic copies of that cube have distance at least $\ell_* - 3\ell_*/(8L) > \ell_*/2 > 1 + 2\eta_*$. Each functional defining one of the three norms reads only bonds with endpoint distances at most $1 + 2\eta_*$. It therefore reads at most one periodic copy. Taking its supremum identifies each torus norm with the corresponding norm of G. Translations and coordinate reflections preserve these norms, including the one-bond shift that reverses an oriented component. The same argument applies to the cell copies.

The image of $v + Q_\beta$ under each wall map lies in $\prod_\mu [c_\mu \ell_c + \ell_c/8, (c_\mu + 1)\ell_c - \ell_c/8]$. For distinct cells, a separating coordinate on the periodic circle gives distance at least $\ell_c/4$: any joining arc contains the two margins of size $\ell_c/8$. Thus supporting endpoints of distinct $\widehat{c}_c$ have stop-coordinate distance at least $\ell_*/(8L)$, which is larger than $1 + 2\eta_*$. Again each external-norm functional reads at most one copy, so

$$|a(t)|_{\mathrm{ext}} \le C_\beta |t|_\infty / e, \qquad v_\eta = \|d\widehat{\mathfrak{a}}\|^2 \le \mathbf{D}. \tag{60}$$

For the second bound, the covering curvature dW is supported in the cube of side $D_\beta^{\mathrm{buf}}/\eta + 2 < n_*$ obtained by enlarging $\eta^{-1}(v + Q_\beta)$. Its periodic translates do not overlap, so periodization preserves its counting norm. Distinct cell curvatures also have disjoint plaquette supports: a common plaquette would have vertices in both endpoint sets at physical distance at most $\eta \le 1$, whereas $\ell_c/4 > 1$. This gives (15) for every admitted placement.

Let P_c be the unit-block centers whose blocks contain an endpoint of a supporting bond of $\widehat{c}_c$, and let P_c^+ be their $(M+1)$-neighborhood. For $y \in P_c$, $y' \in P_{c'}$ choose the corresponding endpoints in $\Delta_y, \Delta_{y'}$. Each endpoint is at distance less than $1/2$ from its center. Consequently

$$d_{\mathrm{sep}} := \min_{c\ne c'} d(P_c, P_{c'}) \ge \ell_*/(8L) - 1 = Mr_*/(8L) - 1, \tag{61}$$

$$|P_c^+| \le p_\beta := (D_\beta^{\mathrm{buf}}/a_* + 2M + 4)^3. \tag{62}$$

For the count, the endpoints have a lift in a cube of side $D_\beta^{\mathrm{buf}}/a_*$. A center adds less than $1/2$ to each coordinate bound, and its neighborhood adds $M + 1$. The resulting open interval has length $D_\beta^{\mathrm{buf}}/a_* + 2M + 3$ and contains fewer than one plus that many integers. Multiply the three counts. If the lattice probe vanishes, all these sets are empty and $d_{\mathrm{sep}} = \infty$.

5. The terminal inputs

We recall the terminal estimates used in the response proof, with the normalizations of [26j, §6 and Appendix A]. They apply to the common trajectory fixed in the introduction. The marked activity estimates, logarithm and complement argument are proved in the following sections.

5.1. **Operators, norms and constants.** At the reference stop $T_* = \eta_*\mathbb{Z}^3/\ell_*\mathbb{Z}^3$, coarse spinors live at $T_*^0 = \mathbb{Z}^3/\ell_*\mathbb{Z}^3$ with weight one; fine chamber indices have weight $\eta_*^3/48$. The block average Q_* and cluster injection J are those of [26b, §2.1 and §4.2]. The fine resolvent and blocked operator are

$$\begin{aligned} F_*(A) &= D_A + \mu_* + b_* J P_*(A) J^*, \qquad S_*(A) = F_*(A)^{-1}, \\ H_*(A) &= b_* S_*(A) J Q_*^T(-A), \qquad D_*(A) = b_* - b_* Q_*(A) J^* H_*(A). \end{aligned} \tag{63}$$

Here $P_* = Q_*^T(-A)Q_*(A)$ on the cluster-averaged space, and $H_*^{\#}$ includes ordinary and covariant Hölder-pair rows. The barred rows are their charge-conjugate partners. The parameter ranges are

$$b_- = 1 - L^{-1} \le b_* \le 1, \qquad \mu_- = L^{-3} < \mu_* \le L^{-1}, \qquad \gamma^F M \ge 16, \qquad \ell_* > 2M + 2.$$

The positive mass $\mu_* = \bar{m} a_*$ is distinct from the running mass shift m_* and from the possibly signed total fine mass.

The domains $\widetilde{R}_j$, $\mathfrak{X}_j$ and $\widetilde{G}_j$ are the background, external and total-connection domains of [26c, §6] and [26f, §2.3]. In particular

$$\mathfrak{X}_j = \{C : |C|_\infty < e_j^{-1+2\varepsilon},\ |\partial C|_\infty < e_j^{-1+3\varepsilon}\}, \qquad \widetilde{R}_j + \mathfrak{X}_j \subset \widetilde{G}_j.$$

The local real-gauge definition of $\widetilde{R}_j$ makes it invariant under real constants. At the stop put

$$\mathcal{W} = \{A_r + C : A_r \in \widetilde{R}_* \text{ real},\ C \in \mathfrak{X}_*\}.$$

On this domain the inverse and field maps are analytic. Their localized majorants, including short-pair Hölder rows, are bounded by $C^F E(y,y')$, where $E(y,y') = \mathrm{e}^{-\gamma^F d(y,y')}$. For long pairs the distance is the minimum of the distances from the two endpoint blocks. All global scalar component row sums of $H_*^{\#}$ and its barred partner are at most C^F.

Localize a fine index in its unit block and a pair at its first point. For an operator O define

$$|O|(y,y') = \sup_{\|v\|_\infty \le 1} \|\mathbf{1}_y O \mathbf{1}_{y'} v\|_\infty,$$

$$\|O\|_\gamma = \max\left\{ \sup_y \sum_{y'} |O|(y,y')\mathrm{e}^{\gamma d(y,y')},\ \sup_{y'} \sum_y |O|(y,y')\mathrm{e}^{\gamma d(y,y')} \right\}.$$

The spinor norm here is the maximum component modulus. For coarse operators, $\|O\|_2 \le 2\|O\|_0$ and $\|O\|_1 \le 4\sum_{y,y'} |O|(y,y')$, where the subscripts 2 and 1 denote operator and trace norm, respectively.

For a direction B, let P_B be the unit blocks containing an endpoint of a bond in its support, put $P_B^+ = \{y : d(y,P_B) \le M+1\}$ and $\epsilon_B = e_*|B|_\infty$. The first coarse derivative has majorant $226(C^F)^2\epsilon_B \sum_{z\in P_B} E(y,z)E(z,y')$. The first field-leg derivative has the analogous two-edge chain through P_B^+, with coefficient $C_H\epsilon_B$ and a sum over at most two endpoint blocks. For directions separated by $M+2$, both mixed derivatives have the sum of the two ordered three-edge chains through $P_B, P_{B'}$, or their enlarged sets for the legs, with coefficient $C_H^2\epsilon_B\epsilon_{B'}$. These precise pointwise estimates are [26j, Proposition A.1]; the response proof uses their support factors as well as their row sums.

We retain their constants:

$$\begin{aligned} \mathcal{S} &= C^F(24/\gamma^F)^3, & h_{\mathrm{L}} &= 6\mathcal{S}, & C_H &= 203M(C^F)^2, \\ C_K &= 226\mathcal{S}^2, & C_K' &= 1328\mathcal{S}^2, & c_\tau &= \min\{1/8, \mu_-/48, (292\mathcal{S})^{-1}\}, \\ g &= (b_-^{-1} + \mu_-^{-1})^{1/2}, & h_G &= 1+g, & g_1 &= b_-^{-1} + 4/\mu_-. \end{aligned}$$

Set $\theta_{\mathrm{CT}} = \mu_-/(12\mathrm{e})$ and

$$\vartheta_0 = \min\{\theta_{\mathrm{CT}}/2, \gamma^F/8, 1/M\}, \qquad \alpha_G = 2b_-^{-1} + 4\mathrm{e}\mu_-^{-1}(24/\theta_{\mathrm{CT}})^3.$$

The real covariance $G = D_*(A_r)^{-1}$, for every real A_r, satisfies $\|G\|_2 \le g^2$ and has weighted component row and column sums at most α_G, with weight $\mathrm{e}^{\vartheta_0 d(u,u')}$. On $\mathcal{W}^+ = \{A + A_1 : A \in \mathcal{W},\ e_*|A_1|_\infty \le c_\tau\}$, the first derivative bound holds with C'_K. If $A = A_r + A_1$, A_r is real, $e_*|A_1|_\infty \le \mu_*/24$ and $F_*(A)$ is invertible, then $D_*(A)$ is invertible and $\|D_*(A)^{-1}\|_2 \le g_1$. These are [26j, (72)–(75)].

The activity space, independent ordinary and pair generators, and their coefficient norm are those of [26f, §2.2]. An activity supported on an M-polymer X has coefficients E_{nm}, integrated with the fine-index weights above and pair weights $(\eta_*^3/48)^2$, and norm $\sum_{n,m} h_1^n h_2^m \|E_{nm}\|/(n!m!)$. The family norm adds the factor $\mathrm{e}^{\kappa d_M(X)}$ and takes the supremum over polymers and the declared connection domain. Here $Md_M(X)$ is the infimum of the lengths of connected finite nets meeting every closed M-cube of X. A net is a finite union of images of closed straight segments in the continuum torus; its length is the sum of their Euclidean lengths, with the infimum also taken over segment representations. A point is allowed as a segment. We use the construction constants $\kappa_0 = 54\log 72$, $K_0 = 72^{54}/36$, $c_{\mathrm{rb}} = 54K_0$ and $\kappa \ge 7\kappa_0 + 7$. The anchored polymer sum is bounded by K_0, and its reblocking bound has constant c_{rb}, as proved in [26f, Appendix B, (79)–(83)]. The fixed constant $C_0 \ge C^F$ bounds the photon kernel rows and the field substitution; in particular at $a_2 = C_0^{-1} e_*^{-1/4+\varepsilon}$,

$$|E(H_*^{\#}\Psi)|_{a_2} \le \|E\|_{h_*}, \qquad |\langle\bar\psi,\psi\rangle|_{a_2} \le 4e_*^{-1/2+2\varepsilon}\ell_*^3, \quad h_* = (e_*^{-1/4}, e_*^{-1/4+\varepsilon}).$$

The scalar constant C_d is the local determinant-activity bound in (68). All constants in this subsection are fixed before the physical charge and are independent of the fine cutoff, twist and placement.

5.2. **The measure identity and Gaussian domination.** We use the exact terminal density of [26j, Theorem B], including its external slot. Its original measure is $\nu^{\theta,a}$ from (39); write $\bar\nu$ for its zero-source twist average and $\bar\mu = \bar\nu/\bar\nu(1)$. The full photon-coordinate map sends an exact fine curvature $F \in F_N = dC^1(C(n_*);\mathbb{R})$ to $(A_*;\zeta_j)_{j<k_*}$. Its Gaussian marginal at the stop is denoted m_*. Every scalar fluctuation coordinate is an orthonormal linear functional of F; every coarse plaquette has row norm at most one. The numbers of coordinates are at most $2\ell_*^3 L^{3(k_*-j)}$ at level j and $3\ell_*^3$ at the stop. The stop Gaussian is nondegenerate on the mean-zero coclosed space.

Set $\chi_*(A) = \mathbf{1}_{\{|dA|_\infty \le p_*\}}$ and

$$\mathfrak{G} = \{|\zeta_j(\Upsilon)| \le p_{0,j}\ (j < k_*),\ |dA_*|_\infty \le p_*\}.$$

For every bounded Borel function Φ of A_*,

$$\nu^{\theta,a}(\mathbf{1}_{\mathfrak{G}}\Phi(A_*)) = \int \Phi(A)\chi_*(A)Y_*^x(A + c_\theta;\mathscr{C})\,\mathrm{d}\mathsf{m}_*(A), \qquad \mathscr{C} = a^{[k_*]}, \quad c_{\theta,\mu} = \theta_\mu/(e_*\ell_*). \tag{64}$$

The density is jointly continuous on the indicated support and is invariant under admissible real gauge changes. Its explicit fermion formula is recalled in section 6.1. The source-free normalized measure satisfies

$$\bar\mu(\mathrm{e}^{\langle f,F\rangle}) \le \mathrm{e}^{\|f\|^2/2} \qquad (f \text{ a real two-cochain}). \tag{65}$$

This is the zero-sector result [26j, Proposition 2.1]. The tower and Borel identity are [26j, Proposition 3.1, Theorem B]; they precede the response argument and do not require its conclusion.

6. The terminal determinant ratio

The argument in this and the following sections is the terminal response proof of [26i, §23]. The free coefficients do not by themselves control the reference Wick products. We must compare the

normalized interacting generating function with its one-cell free factors, including mixed derivatives in different cells. The quantity to be estimated is

$$\Theta_m^{(0)} = \left| \left(\prod_{c \in \mathcal{C}_*} (\partial_{t_c}^m - c_m) \mathring{\Phi} \right)(0) \right|^{1/N_*}, \qquad m = 2, 4, \qquad N_* = |\mathcal{C}_*| = 8L^3, \tag{66}$$

where $c_m = N_0^{(m)}(0)$. The exact relation to Wick products is (53). The proof uses decay between separate insertions, an analytic logarithm for the terminal fermion integral, and a separate estimate of the complement of the small-field set.

6.1. **Analyticity and normalization.** For the marked connections of (57), set

$$C_b = 1 + \sqrt{h}\, C_\beta, \qquad t_x = \frac{e_*^{3\varepsilon}}{C_b}, \qquad t_x^+ = \frac{e_*^{3\varepsilon}}{\sqrt{h}\, C_\beta}, \qquad r = t_x/4. \tag{67}$$

If $C_\beta = 0$, interpret $t_x^+ = \infty$. For t in the polydisc $\mathbb{D}(t_x^+) = \{|t_c| < t_x^+ \text{ for every } c\}$, all the rescaled connections $a(t)^{[j]}$ belong to $\mathfrak{X}_j$ and $e_* |\mathscr{C}(t)|_\infty < e_*^{3\varepsilon}$. Indeed, (60) and $e_* \le e\sqrt{h}$ give $|a(t)|_{\mathrm{ext}} < e_*^{-1+3\varepsilon}$. This bounds the stop sup and first-difference norms by the defining thresholds of $\mathfrak{X}_*$. For $i = k_* - j$ the rescaling multiplies the sup norm by $L^{-i/2}$ and the first differences by $L^{-3i/2}$, while $e_j = L^{-i/2} e_*$. Since $\varepsilon < 1/3$, the rescaled bounds are strictly below $e_j^{-1+2\varepsilon}$ and $e_j^{-1+3\varepsilon}$, respectively.

We first control the determinant activities accumulated along the flow. Recall the local determinant function $E_j^d(X, A)$ of [26b, Proposition 6.1] and set $E_j^{\det}(X, A) = E_j^d(X, A) - E_j^d(X, 0)$. Its analyticity, locality, real gauge invariance, and evenness will all be used below; the relevant bound is

$$|E_j^d(X, A)| \le C_d M^3 \mathrm{e}^{-(\kappa+1) d_M(X)} \qquad (A \in \widetilde{G}_j). \tag{68}$$

For a level-$(j+1)$ bond cochain C, write $C_L(b) = L^{-1/2} C(b/L)$ for its one-step rescaling. Let $\mathfrak{P}_0 = 0$ and

$$\mathfrak{P}_{j+1}(Y, C) = \sum_{\overline{X}^L = LY} \big(\mathfrak{P}_j(X, C_L) + E_j^{\det}(X, C_L) \big). \tag{69}$$

Then $\mathfrak{P}_j(X, \cdot)$ is analytic on $\mathfrak{X}_j$, vanishes at zero, and depends only on bonds meeting X. To see locality under reblocking, a bond meeting X has, after division by L, an endpoint in Y whenever $\overline{X}^L = LY$. Analyticity follows from $C_L \in \mathfrak{X}_j$.

For $d_M(Y) \ge 1$, the reblocking estimate gives

$$\sum_{\overline{X}^L = LY} \mathrm{e}^{-\kappa d_M(X)} \le c_{\mathrm{rb}} L^3 \mathrm{e}^{-L(\kappa - \kappa_0 - 1) d_M(Y)}.$$

With $L(\kappa - \kappa_0 - 1) \ge \kappa$ and $c_{\mathrm{rb}} L^3 \mathrm{e}^{-[L(\kappa-\kappa_0-1)-\kappa]} \le 1/2$, the weighted supremum q_j of $\mathfrak{P}_j$ over $d_M(X) \ge 1$ satisfies $q_{j+1} \le (q_j + 2C_d M^3)/2$. Since $q_0 = 0$,

$$|\mathfrak{P}_j(X, C)| \le 2 C_d M^3 \mathrm{e}^{-\kappa d_M(X)} \qquad (d_M(X) \ge 1). \tag{70}$$

At $j = 0$ the determinant activity includes all 192 chamber components. The primitive bound and reblocking estimate have the same constants at that first step.

The determinant identity [26h, (33)] gives

$$\frac{\det(D^{\mathrm{ch}}[qa] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)} = \exp\left(\sum_X \mathfrak{P}_*(X, a^{[k_*]}) \right) \frac{\det D_*(a^{[k_*]})}{\det D_*(0)}, \tag{71}$$

$$\sum_X \mathfrak{P}_*(X, a^{[k_*]}) = \sum_{j < k_*} \sum_X E_j^{\det}(X, a^{[j]}). \tag{72}$$

Here $a^{[j]} \in \mathfrak{X}_j$ for every j. The fine determinant uses the phase-argument convention of (3): its connection argument is $qa(t) = \sqrt{\eta}\sum_c t_c \widehat{\mathfrak{a}}_c$. For a single cell this is precisely the phase connection $q(t/e)\widehat{a}_c$ in (3). Write $n_j = \ell_j^3$ for the number of unit sites at level j; the spinor dimension is $4n_j$ for $j \geq 1$, and $192n_0$ at the incoming chamber level. The one-step identity reads

$$\det D_{j+1}(a^{[j+1]}) = b^{4n_{j+1}} \det D_j(a^{[j]})/\delta Z_j(a^{[j]}).$$

Divide it by its zero-field version and use [26b, (78)]. The resulting ratio acquires the factor

$$\exp\left(-\sum_X E_j^{\det}(X, a^{[j]})\right)$$

at each step, which proves the first identity by iteration. Summing (69) over Y proves the second: each fine polymer has exactly one reblocked image. The zero-field determinants are nonzero by the real covariance formula. At $j = 0$ the incoming determinant has dimension $192n_0$, the outgoing one dimension $4n_1$; the factor b^{4n_1} cancels in the ratio.

Fix a coarse real photon field A with $\chi_*(A) = 1$, and a real twist θ. The real gauge construction in [26c, (32)]–[26c, (33)] gives a representative A^g with $|A^g|_\infty \leq 2\ell_*^2 p_*$. The curvature bound and [26c, Theorem B] put both $A + c_\theta$ and $A^g + c_\theta$ in Dom_{k_*}. Write

$$\mathscr{A}^0 = H_*^{\mathrm{ph}} A^g, \qquad c_{\theta,\mu} = \frac{\theta_\mu}{e_*\ell_*}, \qquad \mathscr{A} = \mathscr{A}^0 + c_\theta.$$

Summing the three photon kernel estimates in [26c, (14)] bounds each field norm of $\mathscr{A}^0$ by $2C_0\ell_*^2 p_* < e_*^{-\varepsilon}/2$; the strict inequality is the corresponding condition in [26j, (43)]. The local photon domains and their invariance under real constants then give $s\mathscr{A}^0 + c_\theta \in \frac{1}{4}\widetilde{R}_*$ for $0 \leq s \leq 1$. In particular all connections needed below belong to $\mathcal{W}$.

The flow bound [26a, (66)] gives an activity with even Grassmann kernels, with equal barred and unbarred degrees, and

$$\|\mathscr{E}_*\|_*^x \leq 4e_*^{1/4-6\varepsilon}, \qquad |m_*| \leq e_*^{3/4-8\varepsilon}/8. \tag{73}$$

The coefficient norm includes the weights $\eta_*^3/48$ for a fine spinor index and $(\eta_*^3/48)^2$ for a pair index, with all admissible pairs, including pairs inside a cluster. Its polymer factor is $\mathrm{e}^{\kappa d_M(X)}$. For $(A, C) \in \widetilde{R}_* \times \mathfrak{X}_*$ define

$$Z_{\mathrm{in}}(A, C) = \int D\Psi \, \exp\left\{-\langle\bar{\Psi}, D_*(A + C)\Psi\rangle - m_*\langle\bar{\psi}, \psi\rangle + \sum_X \mathscr{E}_*(X; A, C, \psi, \chi)\right\}, \tag{74}$$

where $\psi, \bar{\psi}$ are substituted using $H_*^{\#}(A + C)\Psi$ and their barred partners, and $\chi, \bar{\chi}$ using the corresponding covariant Hölder quotients. The inclusion $\widetilde{R}_* + \mathfrak{X}_* \subset \widetilde{G}_*$ makes all these maps analytic; the activity kernels are analytic in (A, C). Berezin integration is coefficient extraction, and the exponential is its scalar exponential times a finite polynomial in the positive-degree coefficients. Thus Z_{in} is analytic on this two-field domain. Set

$$Z_{\mathrm{tm}}(t) = Z_{\mathrm{in}}(\mathscr{A}, \mathscr{C}(t)), \tag{75}$$

$$f_{\mathrm{int}}(t) = \frac{Z_{\mathrm{tm}}(t)}{\det D_*(c_\theta + \mathscr{C}(t))}. \tag{76}$$

The Berezin measure has one pair for each of the $4\ell_*^3$ coarse spinor coordinates. The two-field analyticity gives analyticity of Z_{tm} on $\mathbb{D}(t_x^+)$.

Proposition 6.1. Suppose

$$4C_0\ell_*^3 g_1 \leq e_*^{-1/4+\varepsilon}, \qquad 4\ell_*^3(2C_K + K_0 + 4) \leq e_*^{-2\varepsilon}, \qquad e_*^{1/4-9\varepsilon} \leq 1/8, \tag{77}$$

$$48e_*^{2\varepsilon} \leq \mu_-. \tag{78}$$

Then f_{int} is analytic and

$$|f_{\mathrm{int}}(t)-1| \le \frac{1}{2}e_*^{1/4-9\varepsilon} \le \frac{1}{16}. \tag{79}$$

In particular Z_{tm} has no zero on the polydisc. The normalized terminal ratio

$$R_{\mathrm{tm}}(t;\theta,A) = \frac{Y_*^{\mathrm{x}}(A+c_\theta;\mathscr{C}(t))}{Y_*^{\mathrm{x}}(A+c_\theta;0)} = \exp\left(\sum_X \mathfrak{P}_*(X,\mathscr{C}(t))\right)\frac{Z_{\mathrm{tm}}(t)}{Z_{\mathrm{tm}}(0)} \tag{80}$$

is nonzero, continuous in (t,θ,A) on $\mathbb{D}(t_x^+)\times\mathbb{R}^3\times\{\chi_*=1\}$, and analytic in t. It has a unique continuous logarithm $\mathcal{G}(t;\theta,A)$ that vanishes at zero, and this logarithm is analytic in t. ■

Proof. Use the covariance

$$C_t = D_*(c_\theta+\mathscr{C}(t))^{-1}.$$

The inverse exists by [26j, (75)], because $e_*|\mathscr{C}(t)|_\infty < e_*^{3\varepsilon} < \mu_*/24$. Each covariance row has sum at most $4\ell_*^3 g_1$. Put $a_w = C_0^{-1}e_*^{-1/4+\varepsilon}$. The first condition in (77) gives $a_w \ge 4\ell_*^3 g_1 \ge 1$.

Separate the quadratic form with covariance C_t from the remaining even interaction V_t. Normalized Gaussian Grassmann integration gives

$$|f_{\mathrm{int}}(t)-1| \le \mathrm{e}^{|V_t|_{a_w}} - 1. \tag{81}$$

To justify this particular norm bound, replace the unbarred generators by $C_t\Psi$ and leave the barred generators fixed. Their row sums are at most a_w, so this substitution maps the a_w coefficient norm into the unit coefficient norm. Integration against the identity covariance has norm one in that latter norm. Apply this to $\mathrm{e}^{V_t}-1$, and use the submultiplicativity of the coefficient norm.

There are three terms in V_t. The difference between the two coarse quadratic forms has entry sum at most $C_K e_*^{1-\varepsilon}\ell_*^3/2$, by [26j, (66)] integrated along $c_\theta+\mathscr{C}(t)+s\mathscr{A}^0$. A quadratic Grassmann form has coefficient norm at most $4a_w^2$ times this block sum, so its contribution is bounded by $2C_K\ell_*^3 e_*^{1/2+\varepsilon}$. The terminal mass estimate and (73) give respectively

$$\frac{1}{2}\ell_*^3 e_*^{1/4-6\varepsilon}, \qquad K_0\ell_*^3 e_*^{1/4-7\varepsilon}.$$

For the activity sum, substitute the terminal field maps in the coefficient norm, then sum $\mathrm{e}^{-\kappa d_M(X)}$ by assigning each polymer one of its M-cubes and using the polymer sum bound K_0. We have used $4e_*^{\varepsilon} \le 1$. Consequently

$$|V_t|_{a_w} \le \ell_*^3(2C_K+K_0+4)e_*^{1/4-7\varepsilon} \le \frac{1}{4}e_*^{1/4-9\varepsilon}.$$

The inequality $\mathrm{e}^x - 1 \le 2x$ proves (79). The denominator in (76) is analytic and nonzero, so f_{int} is analytic.

By [26j, (16)] the terminal density has the form

$$Y_*^{\mathrm{x}}(A';C) = (\det D_*(0))^{-1}\exp\left(-\varepsilon_*^{\mathrm{v}}\ell_*^3 + \sum_X \mathfrak{P}_*(X,C)\right) Z_{\mathrm{in}}(H_*^{\mathrm{ph}}A', C).$$

Its gauge invariance in [26j, Theorem B] permits replacing A by A^g. The photon minimizer preserves real constants. Dividing the density at $C=\mathscr{C}(t)$ by its value at zero proves (80); the quotient also shows that the result is independent of the chosen gauge representative. For continuity, use the ungauged linear map $A+c_\theta \mapsto H_*^{\mathrm{ph}}(A+c_\theta)$ in the displayed formula. The terminal Grassmann integral is analytic in its two field arguments, and its zero-mark denominator is nonzero.

The required logarithm is

$$\mathcal{G}(t;\theta,A) = \int_0^1 \sum_c t_c \frac{\partial_{t_c} R_{\mathrm{tm}}(st;\theta,A)}{R_{\mathrm{tm}}(st;\theta,A)}\,\mathrm{d}s. \tag{82}$$

Local Cauchy formulas show that the derivatives in this integrand are jointly continuous; integration over $[0,1]$ preserves continuity and analyticity. Along a ray the integrand is the logarithmic derivative of $R_{\mathrm{tm}}(st)$. Differentiating the product of this ratio with the exponential of minus the integrated logarithmic derivative gives zero, and its value at $s=0$ is one. Hence $\mathrm{e}^{\mathcal{G}} = R_{\mathrm{tm}}$. Any two continuous logarithms differ by $2\pi i$ times an integer-valued continuous function on the connected polydisc. Their common value at zero proves uniqueness. □

7. The one-cell logarithm and harmonic twists

For a fixed cell c and real twist θ, let

$$g_D^{\theta}(t) = \operatorname{tr}\log\Big(1 + D_*(c_\theta)^{-1}[D_*(c_\theta + t\widehat{c}_c) - D_*(c_\theta)]\Big). \tag{83}$$

The logarithm here is the matrix power series. Set

$$\begin{aligned} C_{\mathrm{fr}} &= 2592 C_d K_0 C_\beta^2 \left(M^2 (D_\beta^{\mathrm{buf}})^3 L^4 h^{-2} + 8M^5 h\right), \\ B_{\mathrm{wr}} &= 16\exp(6) C_d M^3 K_0 p_\beta \mathrm{e}^{-r_*}, \qquad B_1 = 16 g_1 C_K' p_\beta e_*^{2\varepsilon}, \\ B_0 &= C_{\mathrm{fr}} e_*^{4\varepsilon} + B_{\mathrm{wr}} + B_1. \end{aligned} \tag{84}$$

Proposition 7.1. Suppose (77)–(78) hold, $r_* > 31L + 16$, and

$$2g_1 C_K' e_*^{2\varepsilon} \le 1/2, \quad B_0 \le 1/4, \quad \tau_s := c_\tau \ell_* \ge 1, \quad 52 B_1 \mathrm{e}^{-\tau_s} \le e_*^{1/4-9\varepsilon}. \tag{85}$$

The free ratio N_0 has the logarithm

$$\ell_0(t) = \sum_X \mathfrak{P}_*(X, t\widehat{c}_c) + g_D^0(t), \qquad N_0(t) = \mathrm{e}^{\ell_0(t)}, \tag{86}$$

analytic on $|t| < t_x^+$. It is independent of c and of the admitted placement, is even and real on the real axis, and satisfies

$$\left|\sum_X \mathfrak{P}_*(X, t\widehat{c}_c)\right| \le C_{\mathrm{fr}} e_*^{-2\varepsilon} |t|^2 + B_{\mathrm{wr}}, \tag{87}$$

$$|\ell_0(t)| \le B_0 \qquad (|t| \le t_x). \tag{88}$$

The one-cell interacting error $w_c(t) = \mathcal{G}(te_c; \theta, A) - \ell_0(t)$ is analytic, vanishes at zero, and obeys

$$|w_c(t)| \le 3 e_*^{1/4-9\varepsilon} \qquad (|t| < t_x^+). \tag{89}$$

■

Proof. Write $K = D_*(c_\theta + t\widehat{c}_c) - D_*(c_\theta)$. The real covariance and the integrated insertion bounds give

$$\|D_*(c_\theta)^{-1} K\|_2 \le 2 g_1 C_K e_*^{2\varepsilon} \le 1/2, \qquad \|D_*(c_\theta)^{-1} K\|_1 \le 4 g_1 C_K p_\beta e_*^{2\varepsilon}.$$

Thus the series in (83) converges, is analytic, vanishes at zero, and has modulus at most B_1. Triangularizing the matrix identifies its exponential with $\det D_*(c_\theta + t\widehat{c}_c)/\det D_*(c_\theta)$. Combine this with (71). The lattice symmetries carrying one cell probe to another preserve the fine determinant, so the resulting fine ratio is N_0.

The bound (87) requires a cancellation before summing the levels. Put $i = k_* - j$ and $B_j = e^{-1}\widehat{\mathfrak{a}}_c^{[j]}$. Rescaling gives

$$e_j |\partial B_j|_\infty \le a_j^2 a_*^{-3/2} C_\beta. \tag{90}$$

Let $\mathcal{Q}_j$ be the M-cubes containing endpoints of the supporting bonds of B_j. The buffered support cube yields

$$|\mathcal{Q}_j| \le 4\left((D_\beta^{\mathrm{buf}})^3 M^{-3} a_j^{-3} + 8\right), \qquad |\mathcal{Q}_j| \le L^{3i} p_\beta. \tag{91}$$

By locality only polymers containing a cube of $\mathcal{Q}_j$ contribute to (72).

For such a polymer X, put $d = d_M(X)$ and $s_X = M(d+3)$. Suppose first that $2s_X < \ell_j$. All bonds meeting X lie in a coordinate cube of radius $M(d+2)+1$ around a chosen site $\tilde{c}$ of X. The strict inequality $2[M(d+2)+1] + \eta_j < 2s_X < \ell_j$ makes this cube a lift to the lattice. Define a real function there by $\lambda(c) = \sum_\nu B_{j,\nu}(\tilde{c})(c-\tilde{c})_\nu$, and extend it by zero outside. On bonds meeting X, its gradient is the constant value $B_j(\tilde{c})$. If $[\,\cdot\,]_X$ restricts a connection to those bonds, then

$$B_X = [B_j - d\lambda]_X, \qquad |B_X|_\infty \le 3s_X |\partial B_j|_\infty.$$

The last inequality follows by integrating the first differences along coordinate paths inside the lifted cube.

Real gauge invariance must be used before continuing to the complex gauge parameter. On the convex open set where $|[tB_j - zd\lambda]_X|_\infty < e_j^{-1+\varepsilon}$, the analytic function $z \mapsto E_j^d(X, [tB_j - zd\lambda]_X)$ equals $E_j^d(X, tB_j)$ for real z near zero: the unrestricted connection is related to tB_j by a real gauge transformation, and locality permits the restriction. The identity theorem gives this equality throughout that set. In particular it holds at $z = t$ whenever $|t| < \rho_X$, where $\rho_X = e_j^{-1+\varepsilon}/(3s_X|\partial B_j|_\infty)$.

The function $z \mapsto E_j^d(X, zB_X) - E_j^d(X, 0)$ is even, bounded by $2C_d M^3 \mathrm{e}^{-(\kappa+1)d}$ on $|z| < \rho_X$, and has a double zero at zero. Apply the maximum principle after dividing by z^2. For $|t| \ge \rho_X$ the original primitive bound gives the same inequality with the factor $(|t|/\rho_X)^2 \ge 1$. Thus, in both cases,

$$|E_j^{\det}(X, tB_j)| \le 18 C_d M^5 C_\beta^2 (d+3)^2 \mathrm{e}^{-(\kappa+1)d} |t|^2 e_j^{-2\varepsilon} a_j^4 a_*^{-3}. \tag{92}$$

When $\partial B_j = 0$, $B_X = 0$ and this contribution vanishes. This proves the cancellation also at the chamber first step.

Since $(d+3)^2 \mathrm{e}^{-(\kappa+1)d} \le 9\mathrm{e}^{-\kappa_0 d}$, summing (92) over cubes and polymers gives at level j the bound

$$648 C_d K_0 C_\beta^2 |t|^2 e_j^{-2\varepsilon} a_*^{-3} \left(M^2 (D_\beta^{\mathrm{buf}})^3 a_j + 8M^5 a_j^4 \right).$$

For the remaining polymers $2s_X \ge \ell_j$, $d \ge r_* L^i/2 - 3$. The extra exponential in the primitive bound then gives at level j

$$2\exp(6) C_d M^3 K_0 p_\beta L^{3i} \mathrm{e}^{-r_* L^i}.$$

Sum over $i \ge 1$, using $a_j = L^{-i} a_*$, $e_j^{-2\varepsilon} = e_*^{-2\varepsilon} L^{i\varepsilon}$, $\sum_i L^{-i(1-\varepsilon)} \le 1$, and $a_*^{-2} \le L^4 h^{-2}$, $a_* \le h$. For $r_* > 31L + 16$, the second sum is bounded by a constant times e^{-r_*}. The stated definitions of C_{fr} and B_{wr} give (87). Together with $|g_D^0| \le B_1$ this proves (88).

Evenness and reality follow from the corresponding fine determinant symmetries. Both $\ell_0(t)$ and $\ell_0(-t)$ vanish at zero and have exponential $N_0(t)$, so they coincide. For real t, the nonzero real ratio $N_0(t)$ is positive by continuity from one at zero; the normalized logarithm is therefore real. Cell and placement independence follows in the same way from the determinant translation and reflection symmetries.

It remains to control the twist. For $|\operatorname{Im}\theta_\mu| \le \tau_s$ the connections $c_\theta + st\widehat{c}_c$ lie in $\mathcal{W}^+$, and $\|D_*(c_\theta)^{-1}\|_2 \le g_1$. There is a positive strip margin $\ell_*(e_*^{2\varepsilon} - e_*^{3\varepsilon})$: split the imaginary constant connection into a part bounded by c_τ/e_* and a remaining part absorbed into $\mathfrak{X}_*$. The latter has vanishing first differences. The mass condition bounds its total imaginary part by $\mu_*/24$. On this larger open strip, [26j, (73)] gives the same matrix power-series bound $|g_D^\theta| \le B_1$. The function is holomorphic there and 2π-periodic in each twist variable. For real twists, periodicity is conjugation by the periodic gauge factor $\exp(-2\pi i c_\mu/\ell_*)$; the identity theorem extends it to the strip.

We recall the short Fourier estimate to make the size of this suppression explicit. A bounded periodic holomorphic function g in a closed strip of width $\tau \ge 1$ has $|\widehat{g}(m)| \le B\mathrm{e}^{-\tau|m|_1}$: shift each Fourier contour towards $-i\tau\,\mathrm{sgn}(m_\mu)$, first using an interior width and then taking its limit. The

vertical sides cancel by periodicity. The nonconstant Fourier sum is bounded by

$$B\left[\left(\frac{1+\mathrm{e}^{-\tau}}{1-\mathrm{e}^{-\tau}}\right)^{3}-1\right]\le 25B\mathrm{e}^{-\tau}.$$

The last inequality follows by expanding the bracket divided by $\mathrm{e}^{-\tau}$ as a power series with positive coefficients and evaluating at $\mathrm{e}^{-\tau}=\mathrm{e}^{-1}$. Thus the difference between two real values is at most $52B\mathrm{e}^{-\tau}$. Apply this to $g_D^{\theta}(t)$ to obtain

$$|g_D^{\theta}(t)-g_D^{0}(t)|\le 52B_1\mathrm{e}^{-\tau_s}\le e_*^{1/4-9\varepsilon}. \tag{93}$$

Finally the normalized logarithm identity and (76) give

$$w_c(t)=g_D^{\theta}(t)-g_D^{0}(t)+\log f_{\mathrm{int}}(te_c)-\log f_{\mathrm{int}}(0).$$

The principal logarithm of f_{int} is analytic and has modulus at most $e_*^{1/4-9\varepsilon}$. Both sides vanish at zero and exponentiate to the same ratio; uniqueness of the logarithm justifies the identity. Equation (93) proves (89). □

Let $\phi_0(t)=\sum_{i=0}^{4}N_0^{(i)}(0)t^i/i!$. Cauchy's estimate applied to $|N_0(t)-1|\le 2B_0$ on $|t|=t_x=4r$ gives

$$|c_2|\le r^{-2}/16,\qquad |c_4|\le r^{-4}/16,\qquad \|\phi_0-1\|_r\le 1/6,\qquad \|\phi_0\|_r\le 2. \tag{94}$$

The odd coefficients vanish. Expanding $\log N_0$ through degree four gives $\ell_0^{(4)}(0)=c_4-3c_2^2$, the chamber fourth coefficient of [26d, Theorem B].

8. Derivatives in separated cells

To estimate mixed logarithmic derivatives, we use the real covariance $G=D_*(\mathscr{A})^{-1}$. On the coarse Grassmann algebra, write an even element without constant term as

$$F=\sum_{p\ge 2}\frac{1}{p!}\sum_{u_1,\dots,u_p}F_p(u_1,\dots,u_p)\Psi(u_1)\cdots\Psi(u_p),$$

with antisymmetric kernels. For a symmetric weight $w\ge 1$, define

$$\rho_G(F;w)=\sum_{p\ge 2}\frac{h_G^{p}}{(p-1)!}\sup_{u_1}\sum_{u_2,\dots,u_p}|F_p(u_1,\dots,u_p)|w,$$
$$N_G(F;w)=\sum_{p\ge 2}\frac{h_G^{p}}{(p-1)!}\sum_{u_1,\dots,u_p}|F_p(u_1,\dots,u_p)|w. \tag{95}$$

The supremum in ρ_G is taken separately at each degree. The indices include site, spin, and barred or unbarred status; their distance is the distance between the sites. We use

$$w_\delta(u_1,\dots,u_p)=\mathrm{e}^{\vartheta_0\max_{i,j}d(u_i,u_j)},\qquad w_P(u_1,\dots,u_p)=\mathrm{e}^{\vartheta_0\max_i d(u_i,P)}.$$

The constants h_G,ϑ_0,α_G are those of [26j, (74)]. Antisymmetrizing a kernel that is not antisymmetric bounds N_G by the same expression for that kernel. For ρ_G, replace the supremum at the first argument by the largest supremum over all argument positions.

In the fine algebra absorb the terminal mass into each single-cube activity:

$$E^{a}(X;t)=\mathscr{E}_*(X;\mathscr{A},\mathscr{C}(t))-\mathbf{1}_{\{X\text{ is an }M\text{-cube}\}}m_*\langle\bar\psi,\psi\rangle_X.$$

Let $E^{a}_{nm}(X;t)$ denote its fine spinor/pair kernels. The antisymmetric mass kernel has norm $8M^3$: its two fine-index weights multiply $48\eta_*^{-3}$, its diagonal Kronecker factors leave $48(M/\eta_*)^3$ chamber sites,

and there are four spin components and two orders. It follows from (73) that, when $a_w \le e_*^{-1/4+\varepsilon}$ and $(4+M^3/2)e_*^{\varepsilon} \le 1$,

$$\sum_{n,m} \frac{a_w^{n+m}}{n!m!} \|E^a_{nm}(X;t)\| \le w_*^{\mathrm{E}} \mathrm{e}^{-\kappa d_M(X)}, \qquad w_*^{\mathrm{E}} = e_*^{1/4-7\varepsilon} \quad (|t|_\infty \le t_x). \tag{96}$$

Cauchy's formula on the circles of radius $t_x/2$ in one or two distinct mark variables gives, on $|t|_\infty \le t_x/2$, the same bound for $\partial_c E^a_{nm}$ and $\partial_c\partial_{c'} E^a_{nm}$ multiplied respectively by $q_x = 2/t_x$ and q_x^2. The circle integrals precede the finite weighted kernel sums, so the coefficient norm is bounded without replacing it by a supremum of individual coefficients. Locality gives $\partial_c E^a(X;t) = 0$ unless $X \cap P_c \neq \varnothing$.

After substituting the terminal field maps, put

$$\begin{aligned} V(t) &= -\langle \bar\Psi, K(t)\Psi\rangle + \sum_X E^a(X;t)[H_*^{\#}(\mathscr{A}+\mathscr{C}(t))\Psi], \\ K(t) &= D_*(\mathscr{A}+\mathscr{C}(t)) - D_*(\mathscr{A}), \qquad V_0(t) = \sum_X E^a_{00}(X;t), \qquad V_+(t) = V(t) - V_0(t). \end{aligned} \tag{97}$$

It is even and analytic, and V_+ has no constant term. Gaussian integration gives

$$Z_{\mathrm{tm}}(t) = \det D_*(\mathscr{A}) \mathrm{e}^{V_0(t)} \int \mathrm{d}\mu_G \, \mathrm{e}^{V_+(t)}. \tag{98}$$

The choice of the real covariance will allow the exponential weights in the tree expansion.

Proposition 8.1. Suppose

$$5h_{\mathrm{L}}h_G \le e_*^{-1/4+\varepsilon}, \qquad (4+M^3/2)e_*^{\varepsilon} \le 1, \qquad \alpha_G \bar r \le 1/2, \quad \bar r = 4h_G^2 C_K e_*^{2\varepsilon} + 650 K_0 e_*^{1/4-7\varepsilon}. \tag{99}$$

Let $d_{\mathrm{sep}} = \min_{c\neq c'} d(P_c, P_{c'})$ and suppose $d_{\mathrm{sep}} \ge 6M+8$. With $s_F = (24/\gamma^F)^3$ and $C_V^{\mathrm{ins}} = 10^3 C_H^2 s_F^4 K_0$, we have

$$\begin{aligned} \rho_G(V_+(t); w_\delta) &\le \bar r && (|t|_\infty \le t_x), \\ N_G(\partial_c V_+(t); w_{P_c}) &\le q_x C_V^{\mathrm{ins}} h_G^2 p_\beta, \quad \rho_G(\partial_c V_+(t); w_{P_c}) \le q_x C_V^{\mathrm{ins}} h_G^2, \\ N_G(\partial_c\partial_{c'} V_+(t); 1) + |\partial_c\partial_{c'} V_0(t)| &\le q_x^2 C_V^{\mathrm{ins}} h_G^2 p_\beta \mathrm{e}^{-\vartheta_0(d_{\mathrm{sep}}-4M-6)}. \end{aligned} \tag{100}$$

The last two lines hold on $|t|_\infty \le t_x/2$, with $c \neq c'$ in the last line. If the lattice probe vanishes, all marked derivatives vanish. ∎

Proof. We give the kernel summation which preserves the marked supports. For a polymer X, let X^+ be its unit-site one-neighborhood. Every fine index in an activity supported on X has its position set in X^+. This includes both chamber sites in an internal pair and the first site of an external nearest-neighbor pair whose other endpoint lies over X. The polymer diameter estimate gives

$$\vartheta_0 \operatorname{diam}(X^+) \le d_M(X) + 3. \tag{101}$$

The weighted kernel after substituting the field maps is, at degree $p = n+m$,

$$K^X_{nm}(u_1,\dots,u_p) = \int E^a_{nm}(X;t;v_1,\dots,v_p) \prod_{i=1}^p \mathrm{Leg}(v_i;u_i)\,\mathrm{d}v.$$

Here $\mathrm{d}v$ includes the fine weights already specified. Passing from the bidegree expression to a degree-p kernel produces the factor $p!/(n!m!)$; in the tree norms its remaining factor is $ph_G^p/(n!m!)$.

Suppose a weight is bounded by $c_w \prod_i \mathrm{e}^{\vartheta_0 d(u_i,\mathrm{pos}(v_i))}$. Each summed leg contributes at most h_{L}. If one argument u is fixed, its leg contributes at most $h_{\mathrm{L}}\mathrm{e}^{-(\gamma^F/4)d(u,X^+)}$. Thus the full kernel sum is at most $c_w h_{\mathrm{L}}^p \|E^a_{nm}(X;t)\|$, and the sum with one fixed argument retains the extra exponential in $d(u,X^+)$. For w_δ, the triangle inequality between the two farthest arguments and their position sets, followed by (101), permits $c_w = \mathrm{e}^{d_M(X)+3}$.

The needed rooted polymer sum is

$$\sum_X \mathrm{e}^{-(\gamma^F/4)d(u,X^+)}\mathrm{e}^{-(\kappa-1)d_M(X)} \le 32K_0. \tag{102}$$

Indeed the sum of $\mathrm{e}^{-(\gamma^F/4)d(u,\square^+)}$ over M-cubes is at most 32. The neighboring 27 cubes contribute at most 27; a cube j layers away has distance at least $(j-1)M$ and there are at most $(2j+1)^3$ such cubes. Use $\gamma^F M/4 \ge 4$ to sum the remaining geometric tail. Assigning a cube of X and applying the anchored polymer sum K_0 proves (102).

Apply (96) with $a_w = 5h_{\mathrm{L}}h_G$. The factor 5^{-p} absorbs the polynomial degree factors. The preceding full and rooted sums give

$$\rho_G\left(\sum_X E^a(X;t)[H_*^{\#}\Psi]_+;w_\delta\right) \le 650K_0 w_*^{\mathrm{E}}.$$

More generally, for kernels bounded by $\mathcal{B}\mathrm{e}^{-\kappa d_M(X)}$ at that weight, these sums give $650K_0\mathcal{B}$ in ρ_G and $\exp(3)\mathcal{B}\sum_X \mathrm{e}^{-(\kappa-1)d_M(X)}$ in N_G. This follows by summing $\sum_{p\ge2} p5^{-p} = 9/80$ in the rooted case and using $p \le 5^p$ in the full case. The quadratic term contributes at most $4h_G^2C_Ke_*^{2\varepsilon}$, since $\vartheta_0 \le \gamma^F/2$. This proves the first line of (100).

For the first derivative, there are three types of term: the coarse quadratic derivative, the activity-kernel derivative, and a derivative of one leg. The first has respective bounds $8h_G^2C_K\epsilon_c p_\beta$ and $4h_G^2C_K\epsilon_c$ in N_G and ρ_G, with $\epsilon_c = e_*|\widehat{c}_c|_\infty \le q_x$. In its chain through $z \in P_c$, the weight w_{P_c} is absorbed by the exponential distances from the two coarse endpoints to z.

For the kernel derivative, X must meet P_c. The same weight comparison uses that meeting point and (101). The Cauchy bound multiplies $\mathcal{B} = w_*^{\mathrm{E}}$ by q_x, and the sum of polymers meeting P_c is at most K_0p_β. Its N_G bound is $\exp(3)K_0p_\beta q_x w_*^{\mathrm{E}}$ and its ρ_G bound is $650K_0q_xw_*^{\mathrm{E}}$.

For a differentiated leg at position j, the insertion estimate has a chain $w \in \mathrm{pos}(v_j) \to z \in P_c^+ \to u_j$. The weight obeys

$$w_{P_c} \le \mathrm{e}^{d_M(X)+5}\mathrm{e}^{\vartheta_0[d(w,z)+d(z,u_j)]}\prod_{i\ne j}\mathrm{e}^{\vartheta_0 d(u_i,\mathrm{pos}(v_i))}.$$

The extra two in the exponent accounts for $d(P_c^+,P_c) \le M+1$. Summing the differentiated leg and the other legs leaves, before the degree sum,

$$2\mathrm{e}^{d_M(X)+5}h_{\mathrm{L}}^{p-1}C_H\epsilon_c s_F\sum_{z\in P_c^+}\mathrm{e}^{-(\gamma^F/2)d(z,X^+)}\|E^a_{nm}(X;t)\|.$$

There are p choices of the differentiated leg. The kernel norm contributes one further factor p, and $p^2 \le 5^p$ absorbs both. Use (102) with $\gamma^F/2$ in place of $\gamma^F/4$. The result is bounded by $9500C_Hs_FK_0w_*^{\mathrm{E}}p_\beta\epsilon_c$ in N_G.

For ρ_G, if the fixed argument lies on the differentiated leg, split its two-edge exponent as $7\gamma^F/8 = 3\gamma^F/8 + \gamma^F/2$. Summing the intermediate site leaves $s_F\mathrm{e}^{-(\gamma^F/4)d(u,X^+)}$. If the fixed argument lies on another leg, that leg supplies the same rooted exponential, and the differentiated leg contributes $2C_H\epsilon_c s_F^2$. Both cases therefore have the bound $9500C_Hs_F^2K_0w_*^{\mathrm{E}}\epsilon_c$. Here the degree sum is $\sum_{p\ge2}p^25^{-p} = 43/160$. Adding the three types of term, using $w_*^{\mathrm{E}} \le 1/8$, $C_K \le C_Hs_F^2$, and $C_H \ge 2$, proves the second line with $C_V^{\mathrm{ins}} = 10^3C_H^2s_F^4K_0$.

For the mixed derivative, put

$$d_0 = d(P_c^+,P_{c'}^+), \quad d_1 = d(X^+,P_c^+), \quad d_2 = d(X^+,P_{c'}^+).$$

Then $d_0 \ge d_{\mathrm{sep}} - 2M - 2$. The differentiated coarse quadratic term is bounded in N_G by

$$16h_G^2C_H^2s_F^3p_\beta q_x^2\mathrm{e}^{-(\gamma^F/2)d_{\mathrm{sep}}}.$$

For an activity kernel with p legs, there are $(p+1)^2$ placements of the two derivatives. They may act on two legs, the same leg, a kernel and a leg, or both on the kernel. The summed first differentiated legs are bounded by $2C_H\epsilon_c s_F^2 \mathrm{e}^{-(\gamma^F/2)d_1}$ and its c' version. The mixed leg is bounded by

$$2C_H^2\epsilon_c\epsilon_{c'}s_F^3\mathrm{e}^{-(\gamma^F/2)d_0}\big(\mathrm{e}^{-(\gamma^F/2)d_1}+\mathrm{e}^{-(\gamma^F/2)d_2}\big).$$

If a derivative acts on the kernel, its corresponding distance d_i is zero by locality. Normalize the first kernel derivatives by q_x and the mixed one by q_x^2. Each of the four resulting kernel types satisfies (96). Every product-rule term is consequently at most $4q_x^2C_H^2s_F^4h_{\mathrm{L}}^p\|E_{\mathrm{type}}\|\mathcal{D}(X)$, where

$$\mathcal{D}(X)=\mathrm{e}^{-(\gamma^F/2)(d_1+d_2)}+\mathrm{e}^{-(\gamma^F/2)d_0}\big(\mathrm{e}^{-(\gamma^F/2)d_1}+\mathrm{e}^{-(\gamma^F/2)d_2}\big).$$

The triangle inequality through X^+ gives $d_0\le d_1+d_2+Md_M(X)+2M+2$. Use $\vartheta_0\le\gamma^F/8$ and $\vartheta_0 M\le 1$ to obtain

$$\mathcal{D}(X)\mathrm{e}^{-\kappa d_M(X)}\le 2\mathrm{e}^{-\vartheta_0(d_0-2M-2)}\big(\mathrm{e}^{-(\gamma^F/4)d_1}+\mathrm{e}^{-(\gamma^F/4)d_2}\big)\mathrm{e}^{-(\kappa-1)d_M(X)}.$$

Sum the two distance factors over the sites of P_c^+ and $P_{c'}^+$, then use (102). The degree factor $p(p+1)^2$ is at most 5^p for $p\ge 2$. At $p=0$ there is one kernel term with no legs, and the same bound controls $\partial_c\partial_{c'}V_0$. The four kernel types contribute in total at most $256C_H^2s_F^4K_0p_\beta q_x^2\mathrm{e}^{-\vartheta_0(d_{\mathrm{sep}}-4M-4)}$. Adding the quadratic term proves the last line of (100). If the lattice probe is zero, $\mathscr{C}(t)=0$ and there is no mark dependence. □

We next turn these kernel estimates into a bound for the logarithm. The finite-dimensional tree formula is recalled with its proof, since the distance between its two marked vertices is the source of the mixed-cell decay.

For n copies of the generators, let $\Delta^{ij}=\sum_{x,y}G(x,y)\partial_{\bar\Psi^{(j)}(y)}\partial_{\Psi^{(i)}(x)}$ and $\Delta_{\{i,j\}}=\Delta^{ij}+\Delta^{ji}$. The derivatives are left Grassmann derivatives. For a forest T and edge parameters $\tau\in[0,1]^T$, let $s_{ij}^T(\tau)$ be the minimum parameter on the path from i to j, zero if there is no path, and $s_{ii}^T=1$. Write $\Delta[s]=\sum_{i,j}s_{ij}\Delta^{ij}$ and let ϵ_0 extract the constant term. For even elements $F_1,\dots,F_n$ without constant term, define

$$T_n(F_1,\dots,F_n)=\sum_{\text{trees }T}\int_{[0,1]^T}\epsilon_0\left\{\prod_{\ell\in T}\Delta_\ell\mathrm{e}^{\Delta[s^T(\tau)]}\prod_i F_i(\Psi^{(i)})\right\}\mathrm{d}\tau. \tag{103}$$

For $n=1$ this is the Gaussian expectation.

Proposition 8.2. If $V_+(t)$ satisfies (99) and is analytic, then

$$W(t)=V_0(t)+\sum_{n\ge1}\frac{1}{n!}T_n(V_+(t),\dots,V_+(t))$$

is analytic on $\mathbb{D}(t_x)$ and $Z_{\mathrm{tm}}(t)=\det D_*(\mathscr{A})\mathrm{e}^{W(t)}$. For $c\ne c'$ and nonempty unit-site sets P,P',

$$\begin{aligned}|\partial_c\partial_{c'}(W-V_0)|&\le 2N_G(\partial_c\partial_{c'}V_+;1)\\&\quad+2\alpha_GN_G(\partial_cV_+;w_P)\rho_G(\partial_{c'}V_+;w_{P'})\mathrm{e}^{-\vartheta_0d(P,P')}.\end{aligned} \tag{104}$$

■

Proof. The forest interpolation formula for a polynomial H is

$$H(\mathbf{1})=\sum_F\int\left(\prod_{\ell\in F}\partial_{s_\ell}H\right)(s^F(\tau))\,\mathrm{d}\tau.$$

For a partition π of the vertices, keep fixed entries $s_{\mathrm{in}}(i,j)$ inside its blocks and put $X_\pi(u)_{ij} = u$ on pairs joining distinct blocks. The chain rule and the fundamental theorem of calculus give

$$G(X_\pi(u)) = G(X_\pi(0)) + \sum_{\substack{\ell=(i,j)\\ i,j \text{ in distinct blocks of } \pi}} \int_0^u (\partial_{s_\ell} G)(X_\pi(w))\,\mathrm{d}w. \tag{105}$$

When ℓ joins two blocks, merge them and assign the value w to every pair between those two blocks. The interpolation point is unchanged. Begin with singleton blocks, $G = H$ and $u = 1$, and iterate this identity on each integrand. Each derivative merges two components, so after at most $n-1$ steps the procedure ends. The successive parameters are ordered decreasingly; when two vertices merge, their common value is the minimum parameter on their eventual path. For a fixed forest, its edge orderings divide the parameter cube into these ordered simplices. Summing them gives the formula.

The matrix s^F is a Gram matrix of unit vectors: at $u \in [0,1]$, assign to vertex i the basis vector of its component among the edges with $\tau_\ell > u$, and integrate their scalar products in u. This scalar product is the length of the interval on which the two vertices remain connected, namely s^F_{ij}. Wick's formula and the Gram inequality therefore bound a residual monomial of degree p under $\mathrm{e}^{\Delta[s^F]}$ by g^p, where $\|G\|_2 \le g^2$. Indeed, its determinant entries are $\langle \omega_i \otimes \delta_x, \omega_j \otimes G\delta_y\rangle$. These facts apply whether or not the interpolated covariance is invertible.

Let vertex i of a tree have degree k_i and let the chosen homogeneous part of F_i have degree p_i. The edge derivatives leave the factorial $(p_i - k_i)!^{-1}$ and $p_i - k_i$ uncontracted fields at that vertex; terms with $p_i < k_i$ vanish. Wick's Gram bound supplies $g^{p_i - k_i}$. Sum contracted arguments from the leaves to vertex 1, using the weighted row and column bound α_G for every edge. If nonnegative symmetric weights w_i and a distance D satisfy

$$\mathrm{e}^{\vartheta_0 D} \le \prod_i w_i(\text{arguments at } i) \prod_{\ell=(i,j)} \mathrm{e}^{\vartheta_0 d(\text{edge arguments})},$$

this yields $\mathrm{e}^{-\vartheta_0 D}\alpha_G^{n-1}$ times the full kernel sum at vertex 1 and the rooted sums at the other vertices. The number of trees with degrees $k_1, \dots, k_n$ is $(n-2)!/\prod_i (k_i - 1)!$; here $n \ge 2$, $k_i \ge 1$ and $\sum_i k_i = 2n-2$, as in [Pos19, Proposition 4]. Enlarge the sum over degrees and use

$$\sum_{k=1}^{p} \frac{g^{p-k}}{(k-1)!(p-k)!} = \frac{(1+g)^{p-1}}{(p-1)!} \le \frac{h_G^p}{(p-1)!}.$$

We obtain, for $n \ge 2$,

$$|T_n(F_1, \dots, F_n)| \le \mathrm{e}^{-\vartheta_0 D}(n-2)!\alpha_G^{n-1} N_G(F_1; w_1) \prod_{i=2}^{n} \rho_G(F_i; w_i). \tag{106}$$

For $n = 1$ the bound is $N_G(F_1; 1)$. This also explains why the tree norms use $(p-1)!$ rather than $p!$.

Apply the forest formula to the replicated Gaussian expectation of V_+^n. Different forest components contain disjoint generator sets and their even factors commute, so their expectations factor. Grouping by component partitions gives

$$\int \mathrm{d}\mu_G\, V_+^n = \sum_{\{B_1, \dots, B_q\}} \prod_{a=1}^{q} T_{|B_a|}(V_+, \dots, V_+).$$

The exponential formula for partitions identifies the Gaussian integral with the exponential of the connected tree sum. The identity first holds as a formal series in a scalar multiplying V_+. Bound (106) and $\alpha_G \rho_G(V_+; 1) \le 1/2$ make the connected series absolutely convergent up to that scalar equal to one, so the formal identity is an analytic identity there. The tree terms are polynomial in the activity coefficients, and their series converges locally uniformly in t. Thus W is analytic and its derivatives are obtained term by term.

Two derivatives of the symmetric multilinear tree expression either act on the same vertex or on two different vertices. The first terms, including $n = 1$, sum to at most $2N_G(\partial_c \partial_{c'} V_+; 1)$. For the second terms put the two derivatives at vertices 1 and 2. The weights there are $w_P, w_{P'}$, and those at all other vertices are w_δ. Along the unique path from 1 to 2, the distance $d(P, P')$ is at most the sum of the two endpoint distances, all edge distances, and the argument diameters at intermediate vertices. Hence the weight condition for (106) holds with $D = d(P, P')$. The tree sum leaves $\alpha_G N_G(\partial_c V_+; w_P)\rho_G(\partial_{c'} V_+; w_{P'})\mathrm{e}^{-\vartheta_0 d(P,P')}$ times $\sum_{n\ge 2}(\alpha_G \rho_G(V_+; w_\delta))^{n-2} \le 2$. This proves (104). □

For a function f of the marks, let t^T keep the coordinates indexed by T and set all others to zero, and define its component with variable set S by

$$f_S(t) = \sum_{T \subset S} (-1)^{|S\setminus T|} f(t^T).$$

Then $f = \sum_S f_S$, $f_\varnothing = f(0)$, and f_S vanishes if any variable of S vanishes. The latter statement follows by pairing the terms with and without that variable. In particular a variable absent from f makes every component containing it zero. Two coordinate differences can be represented by two integrals of the mixed derivative. Consequently

$$\sup_{|t|_\infty \le R} |f_S(t)| \le 2^{|S|-2} R^2 \sup_{|t|_\infty \le R} |\partial_c \partial_{c'} f(t)| \qquad (c, c' \in S,\ c \ne c'). \tag{107}$$

Proposition 8.3. Assume the factor and mass conditions (77)–(78), the tree conditions (99), and $d_{\mathrm{sep}} \ge 6M + 8$. For $|S| \ge 2$, the normalized logarithm satisfies

$$\sup_{|t|_\infty \le t_x/2} |\mathcal{G}_S(t; \theta, A)| \le \varpi_{\mathrm{sep}} \le \varpi^+_{\mathrm{sep}}, \tag{108}$$

where

$$\begin{aligned} C_W^{\mathrm{mix}} &= 3C_V^{\mathrm{ins}} h_G^2 + 2\alpha_G (C_V^{\mathrm{ins}})^2 h_G^4 + 4\exp(6) C_d M^3 K_0, \\ \varpi_{\mathrm{sep}} &= 2^{N_*} C_W^{\mathrm{mix}} p_\beta \mathrm{e}^{-\vartheta_0 (d_{\mathrm{sep}} - 4M - 6)}, \\ \varpi^+_{\mathrm{sep}} &= 2^{N_*} C_W^{\mathrm{mix}} p_\beta \mathrm{e}^{-\vartheta_0 (M r_* / (8L) - 4M - 7)}. \end{aligned} \tag{109}$$

■

Proof. The tree representation and uniqueness of the normalized logarithm give

$$\mathcal{G}(t) = \sum_X \mathfrak{P}_*(X, \mathscr{C}(t)) + W(t) - W(0).$$

The component of a determinant activity is zero unless X meets every P_c, $c \in S$. Such a polymer has $d_M(X) \ge d_{\mathrm{sep}}/M - 2 \ge 1$. Use (70), sum the polymers anchored at one of at most p_β cubes, and keep $\mathrm{e}^{-2d_M(X)}$ from $\kappa - \kappa_0 \ge 2$. Its total contribution is at most $2^{|S|} 2\exp(4) C_d M^3 K_0 p_\beta \mathrm{e}^{-\vartheta_0 d_{\mathrm{sep}}}$.

Apply (104) with $P = P_c$, $P' = P_{c'}$ and (100). The bound for $\partial_c \partial_{c'} W$ is

$$q_x^2 p_\beta \left[2C_V^{\mathrm{ins}} h_G^2 \mathrm{e}^{-\vartheta_0 (d_{\mathrm{sep}} - 4M - 6)} + 2\alpha_G (C_V^{\mathrm{ins}})^2 h_G^4 \mathrm{e}^{-\vartheta_0 d_{\mathrm{sep}}} \right].$$

In (107), take $R = t_x/2$ and use $q_x R = 1$. Add the determinant contribution and $|S| \le N_*$ to obtain the first inequality. The placement estimate $d_{\mathrm{sep}} \ge M r_*/(8L) - 1$ proves the second. When the lattice probe is zero, $R_{\mathrm{tm}} = 1$, $\mathcal{G} = 0$ and the first separation error is interpreted as zero. □

9. The positive small-field measure and its complement

[26j, Proposition 3.1] identifies F_N with $(A_*;\zeta_j)_{j<k_*}$. Their scalar coordinates have the form $\zeta_j(\Upsilon)=\langle\phi_{j,\Upsilon},F^{\mathrm{u}}\rangle$ with orthonormal $\phi_{j,\Upsilon}$; each coarse plaquette coordinate $(dA_*)(p)$ has the form $\langle\phi_p,F^{\mathrm{u}}\rangle$ with $\|\phi_p\|\le 1$. There are at most $2\ell_*^3L^{3(k_*-j)}$ coordinates at level j and $3\ell_*^3$ coarse plaquettes. The stop Gaussian dm_* is nondegenerate on the real mean-zero coclosed coarse fields. The row and count statements are [26j, (13)] and [26j, (14)]; the Gaussian assertion is [26j, (12)].

Let $\mathfrak{G}$ require $|\zeta_j(\Upsilon)|\le p_{0,j}$ and $|dA_*|_\infty\le p_*$, and let $\mathfrak{G}^-$ decrease each threshold by one. Put

$$\begin{aligned}\varpi_0&=18\ell_*^3\mathrm{e}^{-p_{0*}/2}+6\ell_*^3\mathrm{e}^{-(p_*-1)^2/(2C_0^2)},\\ t_2&=e\,\min\Big\{N_*^{-1/2},\,C_\beta^{-1},\,(2\sqrt{N_*\mathbf{D}})^{-1}\Big\},\\ \varpi_L&=\sqrt{\varpi_0}\,\mathrm{e}^{\mathbf{D}/2},\qquad M_L=2+\mathrm{e}^{\mathbf{D}/2},\\ \lambda_L&=(\varpi_L^{1/3}M_L^{2/3})^{1/N_*}5(2N_*r/t_2)^4.\end{aligned}\tag{110}$$

A reciprocal with zero denominator is interpreted as $+\infty$. The number $\mathbf{D}$ is the uniform probe energy bound, so $v_\eta\le\mathbf{D}$.

Proposition 9.1. Assume the hypotheses of propositions 7.1 and 8.3 and the tail conditions $v_*\ge 8$ and $p_0v_*^{p_0-1}\ge 28$. For a fixed $0<s<1$, set $\bar\kappa_1=6e_*^{1/4-9\varepsilon}$. Assume

$$2\varpi_{\mathrm{sep}}^{+}\le\bar\kappa_1^{N_*},\quad N_*\bar\kappa_1^{1-s}\le 1/2,\quad \varpi_0\le 1/4,\quad 8N_*t_2\le t_x.\tag{111}$$

There is a finite positive measure v_{sf} on $[-\pi,\pi]^3\times\{\chi_*=1\}$, of mass in $[1-\varpi_0,1]$, such that

$$\frac{\bar\nu^{a(t)}(\mathfrak{G})}{\bar\nu(F_N)}=\int \mathrm{e}^{\mathcal{G}(t;\theta,A)}\,v_{\mathrm{sf}}(\mathrm{d}\theta\,\mathrm{d}A).\tag{112}$$

The normalized complement $L(t)=\bar\nu^{a(t)}(F_N\setminus\mathfrak{G})/\bar\nu(F_N)$ is entire. Its fourth jet in each variable has coefficient norm at radius r at most $\lambda_L^{N_*}$. ∎

Proof. First, the union bound and [26j, (7)] give

$$\bar\mu(F_N\setminus\mathfrak{G}^-)\le\varpi_0.\tag{113}$$

For $i=k_*-j\ge 1$, write $v_j=v_*+i(\log L)/2$. The scale conditions $v_*\ge 8$, $p_0v_*^{p_0-1}\ge 28$ give $p_{0,j}\ge p_{0*}+14i\log L$. Since $(p_{0,j}-1)^2\ge p_{0,j}$, the individual tail is at most $2\mathrm{e}^{-p_{0*}/2}L^{-7i}$. Multiply by the coordinate count $2\ell_*^3L^{3i}$ and sum L^{-4i}. The $3\ell_*^3$ coarse plaquettes give the second term in (110). These estimates yield (113), with room in the coefficient 18.

The exact density identity defines

$$v_{\mathrm{sf}}(\mathrm{d}\theta\,\mathrm{d}A)=\frac{\chi_*(A)Y_*^{\mathrm{x}}(A+c_\theta;0)}{\bar\nu(F_N)}\,\frac{\mathrm{d}\theta}{(2\pi)^3}\mathsf{dm}_*(A).\tag{114}$$

We check positivity of this density, which does not follow from the formal Grassmann representation. For real marks, the original fine law $\nu^{\theta,a(t)}$ is nonnegative. The identity [26j, (21)] says, for every nonnegative bounded Borel Φ,

$$\int\Phi(A)\chi_*(A)Y_*^{\mathrm{x}}(A+c_\theta;\mathscr{C}(t))\,\mathsf{dm}_*(A)=\nu^{\theta,a(t)}(\mathbf{1}_{\mathfrak{G}}\Phi(A_*))\ge 0.$$

The terminal density is continuous. On the interior of the compact convex set $\{\chi_*=1\}=\{|dA|_\infty\le p_*\}$, a nonzero imaginary part or negative real part would persist on a ball of positive Gaussian measure, contradicting this identity. At the boundary, approach along uA, $u\uparrow 1$; zero is an interior point. Thus the density is nonnegative everywhere on this set. Proposition 6.1 makes it nonzero there, hence strictly positive. This reasoning also proves $\mathcal{G}(t;\theta,A)\in\mathbb{R}$ for real t: its exponential is a positive ratio and its continuous value at zero is zero.

Integration of the exact terminal identity over θ, followed by (80), proves (112). At $t = 0$ the mass is $\bar{\mu}(\mathfrak{G})$, which lies in $[1 - \varpi_0, 1]$. All integrals here are well-defined Borel integrals: the density is continuous on the compact support and zero elsewhere.

For real $t = ez$ with $|t|_\infty \le t_2$, put $G_z = \sum_c z_c d\widehat{\mathfrak{a}}_c$. Disjoint supports imply $\|G_z\|^2 = v_\eta |z|_2^2 \le 1/4$. The real source shift gives

$$L(ez) = \mathrm{e}^{-v_\eta |z|_2^2/2} \bar{\mu}\left(\mathbf{1}_{(F_N \setminus \mathfrak{G}) + G_z} \mathrm{e}^{\langle G_z, F^{\mathrm{u}} \rangle}\right).$$

Every photon coordinate changes by at most $\|G_z\| \le 1$ under the shift. Thus the shifted complement is contained in $F_N \setminus \mathfrak{G}^-$. Cauchy–Schwarz, (113), and Gaussian domination at $2G_z$ give

$$0 \le L(t) \le \sqrt{\varpi_0}\, \mathrm{e}^{v_\eta |z|_2^2/2} \le \varpi_L. \tag{115}$$

This is the only place where the lowered thresholds of $\mathfrak{G}^-$ are needed.

The source-shift identity also gives, for complex marks,

$$|\mathring{\Phi}(t)| \le \exp\left(\frac{v_\eta}{2e^2} \sum_c (\operatorname{Im} t_c)^2\right).$$

It follows directly by bounding the expectation of the real exponential and canceling its real quadratic normalization. On $|t|_\infty \le t_2$ this is at most $\mathrm{e}^{\mathbf{D}/2}$. The small-field integral has modulus at most 2. Indeed, evenness and the double zero of ℓ_0 give $|\ell_0(t)| \le B_0 |t|^2 / t_x^2$, while the zero of w_c gives $|w_c(t)| \le (3/8)|t|/t_x$. Their sum over cells at $t_2 \le t_x/(8N_*)$ is at most $1/(256N_*) + 3/64$. The remaining logarithm components sum to at most $2^{N_*} \varpi_{\mathrm{sep}}^{+} \le (2\bar{\kappa}_1)^{N_*}/2 \le N_*^{-N_*}/2$, by (111). Thus $|\mathcal{G}| \le 1/2$ and $|\mathrm{e}^{\mathcal{G}}| \le 2$. Since L is the difference between the full generating function and the small-field integral, $|L| \le M_L$ on that polydisc. It is entire because the finite determinant density is entire and is bounded on compact mark sets uniformly in the real photon field and real twist. The modulus bound in the external-response section supplies the required Gaussian integrable majorant also for either Borel event.

We use the following two-constants estimate. Let $n \ge 1$ be an integer. If a holomorphic function on a polydisc in $\mathbb{C}^n$ of radius R has modulus at most M, and at most $\delta \le M$ on its real points, then its modulus is at most $\delta^{1/3} M^{2/3}$ on the polydisc of radius $R/(2n)$. On the upper half of the unit disc the harmonic function $\omega(z) = \frac{2}{\pi} \arg[(1+z)/(1-z)]$ is zero on the diameter and one on the semicircle. Comparing $\log |f(R'z)|$ with $(1-\omega)\log\delta + \omega \log M$, for $R' < R$, the maximum principle gives the same inequality in the half disc. To handle its endpoints, subtract $\epsilon \log[4/|1-z^2|]$ before applying the principle, then let $\epsilon \downarrow 0$. Its harmonic conjugate gives a holomorphic function to which the modulus principle applies. On $|z| \le q$, $\omega(z) \le (4/\pi) \arctan q$; reflection treats the lower half disc. Let $R' \uparrow R$. Apply this one-variable estimate successively with $q = 1/(2n)$. The remaining exponent of δ is at least $(1 - \frac{4}{\pi} \arctan(1/(2n)))^n \ge 1 - 2/\pi > 1/3$, proving the assertion. The case $\delta = 0$ follows by the identity theorem, and $M = 0$ is immediate.

Apply it to L, with $n = N_*$ and $R = t_2$. Cauchy's coefficient bound gives

$$|L_\alpha| \le \varpi_L^{1/3} M_L^{2/3} (2N_*/t_2)^{|\alpha|}.$$

In the fourth jet, every exponent is at most four. Since $y = 2N_* r/t_2 \ge 1$,

$$\sum_{\alpha_c \le 4} |L_\alpha| r^{|\alpha|} \le \varpi_L^{1/3} M_L^{2/3} (1 + y + \cdots + y^4)^{N_*} \le \lambda_L^{N_*}.$$

Reality on real marks gives real jet coefficients. □

10. The fourth-order cancellation

Work in the finite ring $\mathcal{J} = \mathbb{C}[t_c : c \in \mathcal{C}_*]/(t_c^5 : c \in \mathcal{C}_*)$, with coefficient norm $\|P\|_r = \sum_{\alpha_c \le 4} |P_\alpha| r^{|\alpha|}$. This norm is submultiplicative, since discarding terms with an exponent at least five can only decrease the coefficient sum. The jet map is a ring homomorphism; it keeps degree at most

four in each variable. For $S=\{c\}$ set $k_S=\mathrm{e}^{w_c}-1$ in this ring, and for $|S|\ge 2$ set $k_S=\mathrm{e}^{\mathcal{G}_S}-1$. Every monomial of k_S uses every variable of S and no other variable.

For a component f_S bounded on a polydisc of radius R, Cauchy's estimate at a smaller radius $R'\le R/2$ gives

$$\|f_S\|_{R'}\le \sup|f_S|\left(\sum_{j=1}^{4}(R'/R)^j\right)^{|S|}\le \sup|f_S|.$$

Its monomials contain each variable of S because it vanishes on every corresponding coordinate hyperplane. Use this with $R=t_x/2$, $R'=r$, and use the one-cell estimate on radius t_x, to obtain

$$\|k_{\{c\}}\|_r\le \bar\kappa_1,\qquad \|k_S\|_r\le 2\varpi^{+}_{\mathrm{sep}}\le \bar\kappa_1^{N_*}\quad (|S|\ge 2). \tag{116}$$

We used $\|w_c\|_r\le 3e_*^{1/4-9\varepsilon}\le 3/8$ and $\|\mathrm{e}^P-1\|_r\le 2\|P\|_r$ for $\|P\|_r\le 1/2$. The coefficients are continuous in (θ,A): express the logarithm coefficients by Cauchy integrals on a fixed small torus, and then take the finite polynomial expressions in those coefficients. They are real because the logarithm is real at real marks.

In $\mathcal{J}$ the normalized generating function has the exact representation

$$\mathring{\Phi}=\prod_c\phi_0(t_c)\int\prod_{\varnothing\ne S\subset\mathcal{C}_*}(1+k_S)\,v_{\mathrm{sf}}(\mathrm{d}\theta\,\mathrm{d}A)+L^{[4]}. \tag{117}$$

To justify taking the jet inside the integral, use joint continuity, holomorphy, and $|\mathrm{e}^{\mathcal{G}}|\le 2$ on a fixed polydisc of radius at most t_2. Cauchy's coefficient formulas and Fubini then commute every required coefficient with the finite measure integral. Decompose $\mathcal{G}$ into its variable-set components; its singleton components are $\ell_0(t_c)+w_c(t_c)$. Exponentiation, the jet homomorphism, and $\mathrm{e}^{\ell_0}=N_0$ prove the representation.

Theorem B. Use the fixed construction, common trajectory and terminal inputs of section 5. Suppose the factor and mass conditions (77)–(78), the one-cell conditions (85), the tree conditions (99), and the jet conditions (111) hold. Assume also $r_*>31L+16$, $Mr_*/(8L)-1\ge 6M+8$, $v_*\ge 8$, $p_0v_*^{p_0-1}\ge 28$, and $\lambda_L\le\bar\kappa_1^s$, where $0<s<1$. For $m=2,4$,

$$\Theta_m^{(0)}\le m!r^{-m}(2\exp(1)\bar\kappa_1^s+\lambda_L)\le 8m!r^{-m}\bar\kappa_1^s. \tag{118}$$

With $\mathfrak{e}=s(1/4-9\varepsilon)-12\varepsilon>0$,

$$\begin{aligned}\Theta_4^{(0)}&\le 3\cdot 10^5C_b^4e_*^{\mathfrak{e}}\le 3\cdot 10^5C_b^4h^{1/8}e^{\mathfrak{e}},\\ (1+|c_2|)\Theta_2^{(0)}&\le 3\cdot 10^4C_b^4e_*^{\mathfrak{e}}\le 3\cdot 10^4C_b^4h^{1/8}e^{\mathfrak{e}}.\end{aligned} \tag{119}$$

The constants are independent of the ultraviolet cutoff and the admitted placement. The terminal estimates used before harmonic averaging hold uniformly in the real twist. ■□

Proof. On the one-variable jet space define $\ell_c(t_c^j)=m!\mathbf{1}_{\{j=m\}}-c_m\mathbf{1}_{\{j=0\}}$ and let $\ell=\bigotimes_c\ell_c$. Then $(\Theta_m^{(0)})^{N_*}=|\ell(\mathring{\Phi})|$. The free jet satisfies $\ell_c(\phi_0)=0$. From (94), $|\ell(P)|\le(m!r^{-m})^{N_*}\|P\|_r$. These statements follow by evaluating the functionals on monomials with every exponent in $\{0,m\}$.

Expand the product in (117) over families $\mathcal{Y}$ of nonempty cell sets. If $\mathcal{Y}$ fails to cover a cell c, its error factors do not contain t_c; the factor $\phi_0(t_c)$ is therefore annihilated by ℓ_c. Only covering families contribute. Assign cost one to each singleton and cost N_* to each larger set, and let $a(\mathcal{Y})$ be the sum of these costs. A covering family has $a(\mathcal{Y})\ge N_*$: either it contains a larger set or it contains every singleton. By (116) its coefficient norm is at most $\bar\kappa_1^{a(\mathcal{Y})}$. The free factors have norm at most 2^{N_*}. Keep $\bar\kappa_1^{sN_*}$ from the cost and sum the rest over all families. This sum is

$$(1+\bar\kappa_1^{1-s})^{N_*}(1+\bar\kappa_1^{(1-s)N_*})^{2^{N_*}-1-N_*}\le\exp\left(N_*\bar\kappa_1^{1-s}+2^{N_*}\bar\kappa_1^{(1-s)N_*}\right)\le\exp(1).$$

The last step uses $N_*\bar\kappa_1^{1-s}\le 1/2$. The positive small-field measure has mass at most one. Its total contribution is thus bounded by $(m!r^{-m})^{N_*}\exp(1)(2\bar\kappa_1^s)^{N_*}$. The complement contributes at most $(m!r^{-m}\lambda_L)^{N_*}$. Taking the N_*th root proves (118).

Since $r^{-4}=256C_b^4e_*^{-12\varepsilon}$ and $\bar\kappa_1^s\le 6e_*^{s(1/4-9\varepsilon)}$, the $m=4$ bound has coefficient at most $8\cdot24\cdot1536<3\cdot10^5$. For $m=2$, $1+|c_2|\le(17/16)r^{-2}$, so its coefficient is at most $17\cdot1536<3\cdot10^4$. Finally $0<\mathfrak{e}<1/4$ and $e_*\le e\sqrt{h}$ imply $e_*^{\mathfrak{e}}\le h^{1/8}e^{\mathfrak{e}}$. The scale and probe bounds entering this proof are uniform in the stated choices. □

11. A positive charge interval

We now verify that the response conditions admit a common interval of charges, independent of the ultraviolet cutoff. In addition to (77), (78), (85), (99), and (111), require $r_*>31L+16$, $v_*\ge 8$, $p_0v_*^{p_0-1}\ge 28$, and

$$Mr_*/(8L)-1\ge 6M+8,\qquad \lambda_L\le\bar\kappa_1^s, \tag{120}$$

$$D_\beta^{\mathrm{buf}}/2+1/2\le 3\ell_c/8,\qquad D_\beta^{\mathrm{buf}}+3\le\ell_c. \tag{121}$$

The construction of [26a, §6.1] fixes an odd $L\ge13$, a power M of L with $M\ge L^2$, a localization power $r_{\mathrm{loc}}\ge2$, and cutoff powers $p_0\ge2$, $p>2p_0$. It also fixes $0<\varepsilon\le1/200$ and $0<s<1$ such that

$$\mathfrak{e}=s(1/4-9\varepsilon)-12\varepsilon>0. \tag{122}$$

For the construction fixed in the introduction, take $s=1/2$; then $\mathfrak{e}\ge17/400$.

Proposition 11.1. For the fixed construction constants, $\bar m$, and probe β, there is $e_\sharp(\beta)>0$ such that all response conditions hold for $0<e\le e_\sharp(\beta)$ and every $N\ge N_{\mathrm{ref}}(e)$. The threshold depends on the probe only through $C_\beta,\mathbf{D},D_\beta^{\mathrm{buf}}$. It is independent of N, the twist, and the admitted placement. All centered placements are admitted in this range. ■

Proof. Write $v_K=-\log(e\sqrt{h})$ and $r_K=\min\{L^i:L^i\ge v_K^{r_{\mathrm{loc}}}\}$. Then $R^*(e)=hMr_K/L$ and $\ell_c=R^*(e)/(2L)$. [26a, Proposition 6.1] and [26a, (53)] give a finite $N_{\mathrm{ref}}(e)$.

For $N\ge N_{\mathrm{ref}}(e)$ the stop level is $K-1$ or $K-2$ by [26a, (54)]. In particular its scales are independent of N and satisfy

$$\begin{gathered}v_K\le v_*,\quad e_*=\mathrm{e}^{-v_*},\quad a_*\in\{h/L,h/L^2\},\\ v_*^{r_{\mathrm{loc}}}\le r_*,\quad Mv_*^{r_{\mathrm{loc}}}\le\ell_*<LMv_*^{r_{\mathrm{loc}}},\quad p_{0*}=v_*^{p_0},\quad p_*=v_*^p,\\ p_\beta\le p_\beta^+:=(D_\beta^{\mathrm{buf}}L^2/h+2M+4)^3,\qquad \ell_c\ge hMv_K^{r_{\mathrm{loc}}}/(2L^2).\end{gathered} \tag{123}$$

The strict wrapping condition $r_*>31L+16$ already follows from [26a, (50)] whenever $e\sqrt{h}\le e_1$. Thus every response condition is a condition on e alone once the construction and probe are fixed.

A bound $Av^d\mathrm{e}^{-\gamma v}$ with $A,d\ge0$ and $\gamma>0$ tends to zero, uniformly on a sufficiently large half-line. Split the exponential in half and use $v^d\mathrm{e}^{-\gamma v/2}\le(2d/(\gamma\exp(1)))^d$. Likewise, cv^2 exceeds $bv+\log^+A$ on a sufficiently large half-line for $c>0$. We apply these two inequalities to the following groups of conditions.

The local tree smallness and terminal factor conditions follow from exponential decay with positive rates ε, 2ε, $1/4-\varepsilon$, and $1/4-9\varepsilon$; the volume factors are bounded by $(LM)^3v_*^{3r_{\mathrm{loc}}}$. For $\bar r$, use $1/4-7\varepsilon>2\varepsilon$. The complex mass and coarse determinant conditions similarly have rate 2ε. The support count is bounded by p_β^+. All three summands of B_0 tend to zero: their rates are 4ε, 2ε, and the quadratic decay $\mathrm{e}^{-r_*}\le\mathrm{e}^{-v_*^2}$. Also $\tau_s\ge c_\tau Mv_*^2$, so both $\tau_s\ge1$ and $52B_1\mathrm{e}^{-\tau_s}\le e_*^{1/4-9\varepsilon}$ hold once v_* is large.

The separation condition follows from $r_*\ge v_*^2$. Its error obeys

$$2\varpi_{\mathrm{sep}}^+\le 2^{N_*+1}C_W^{\mathrm{mix}}p_\beta^+\mathrm{e}^{\vartheta_0(4M+7)}\mathrm{e}^{-\vartheta_0Mv_*^2/(8L)}.$$

Since $\bar\kappa_1 \ge \mathrm{e}^{-v_*/4}$, quadratic decay makes this at most $\bar\kappa_1^{N_*}$ for all sufficiently large v_*. The family summation condition $N_*\bar\kappa_1^{1-s} \le 1/2$ uses the positive rate $(1-s)(1/4-9\varepsilon)$.

For the photon complement, the inequalities $v_* \ge 8$ and $p_0 v_*^{p_0-1} \ge 28$ hold on a sufficiently large half-line, since $p_0 \ge 2$. Moreover $p > 2p_0$ implies $(p_*-1)^2 \ge C_0^2 p_{0*}$ once v_* is large. Therefore

$$\varpi_0 \le 24(LM)^3 v_*^{3r_{\mathrm{loc}}} \mathrm{e}^{-v_*^2/2}. \tag{124}$$

This is eventually at most $1/4$. Moreover t_2/e is fixed and positive, while e is between $(L/h)^{1/2}e_*$ and $Lh^{-1/2}e_*$. Hence $8N_*t_2/t_x$ is bounded by a fixed constant times $e_*^{1-3\varepsilon}$, which tends to zero.

The potentially largest loss is the fourth derivative radius in λ_L. From (124), the definition of λ_L, and $r/t_2 \le C\mathrm{e}^{v_*}$, we obtain

$$\lambda_L \le A_8 \exp\left(-\frac{v_*^2}{12N_*} + \left(\frac{r_{\mathrm{loc}}}{2N_*} + 4\right) v_*\right) \tag{125}$$

for a finite $A_8 > 0$ depending only on the fixed constants and probe bounds. For clarity one can take

$$A_8 = [24(LM)^3]^{1/(6N_*)} \mathrm{e}^{\mathbf{D}/(6N_*)} M_L^{2/(3N_*)} 5 \left[\frac{N_*\sqrt{h}}{2C_b\sqrt{L}}(t_2/e)^{-1}\right]^4.$$

If $v_*^2/(12N_*) \ge (r_{\mathrm{loc}}/(2N_*) + 17/4)v_* + \log^+ A_8$, then $\lambda_L \le \mathrm{e}^{-v_*/4} \le \bar\kappa_1^s$. Thus the complex continuation and the fourth derivatives preserve a positive small-charge interval. Finally the two placement inequalities hold when $\ell_c \ge (4/3)D_\beta^{\mathrm{buf}} + 3$; this follows from (123) as $v_K \to \infty$.

Choose a finite V beyond all the half-lines just specified and put $e_V = h^{-1/2}\min(e_1, \mathrm{e}^{-V}) > 0$. Every $0 < e \le e_V$ has $v_* \ge v_K \ge V$ and satisfies every response condition. Define $e_\sharp$ as one half of the supremum of the charges $e' \le e_1 h^{-1/2}$ for which all conditions hold for every $0 < e \le e'$. Its defining set contains e_V, so $e_\sharp \ge e_V/2 > 0$. For $e \le e_\sharp$, a member of the defining set exceeds $e_\sharp$, which ensures that all inequalities hold at e even if the scale functions have jumps. This avoids assuming monotonicity of the rounded stop scales.

The scale identities depend on e, h and the fixed construction constants; the probe enters through $C_\beta, \mathbf{D}, D_\beta^{\mathrm{buf}}$. They do not involve the twist or placement, and they are unchanged by the coordinate reflections used to form the cell probes. Centered placements satisfy the admission inequality by (121), since their centers differ from the cell center by at most $\eta/2 \le 1/2$. The same conditions give $D_\beta^{\mathrm{buf}} + 3\eta \le \ell_c$ and $R^*(e) \ge D_\beta + 1$. □

Proof of theorem A. The base construction gives the positive physical ceiling

$$e_G = \begin{cases} e_1, & h = 1, \\ \min\{e_1/\sqrt{h},\, 1/(\sqrt{h}\log h)\}, & h > 1, \end{cases}$$

as in [26g, (31)]. On this interval the common pair and all upstream terminal estimates apply. Choose $e_{\mathrm{RR}} = \min\{e_G, e_\sharp(\beta)\}$, with the threshold of proposition 11.1, at $s = 1/2$. That proposition verifies the additional inequalities of theorem B for every smaller positive charge and every $N \ge N_{\mathrm{ref}}(e)$, and admits every centered placement. The bounds (119) give (4) with the stated common constant. The exact unfolding identifies the normalized density with the zero-flux law, proposition 3.1 proves entireness, and (53) proves (5). Its differentiation proof establishes the required integrability. □

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